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ZERO-KELVIN COMPRESSION ISOTHERMS OF THE ELEMENTS $1 \leq Z \leq 90$ TO 100 GPa (1 MEGABAR)

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ABSTRACT

Most of the chemical elements have now been compressed close to or above 100 GPa (1 Mbar) pressure in diamond-anvil cells and the pressure-volume room-temperature isotherms have been measured. We collect these data and use simple lattice-dynamics models to reduce the isotherms to zero Kelvin.

We have extended the published work by making new diamond-anvil-cell measurements on Cr and Rh, and by conducting density-functional calculations on the elements Po, At, Rn, Fr, Ra, and Ac. The 0 K data are tabulated for all elements $1 \leq Z \leq 90$ and $0 \leq P \leq 100$ GPa.

These data are useful for generating wide-range equation of state models and for studying the stability of chemical compounds at high pressure (“Megabar chemistry”). The tables presented here are intended to

be reference thermodynamic tables for use in high-pressure research. Further experimental and theoretical work will be needed to extend the tables to higher pressure and to improve accuracy.

1. INTRODUCTION

Thermophysical tables are a very useful component of modern research in the physical sciences. Most of these tables, such as heats of formation, equations of state, heat capacities, thermal expansion, resistivities, etc. are tabulated at 1 GPa pressure or less ¹⁻⁵. Very-high-pressure data in the 100 GPa range are still comparatively scarce, although tabulations of high-pressure properties are found in a few volumes of shock-wave physics ^{6,7}.

In developing wide-range equation of state (EOS) models, we need to include both temperature and pressure variables. In the $P - T$ plane, the $T = 0$ compression isotherm is a vertical line, and the $P = 1$ atm expansion isobar is a horizontal line. The EOS is then determined from experiments and theories in the whole $P - T$ space, with these lines as the boundaries.

In this paper we have generated data for the zero-Kelvin compression isotherms of the chemical elements. The data range from hydrogen to thorium and from zero to 100 GPa. The purpose of this work is to increase the visibility of the pressure variable in physical science research and to encourage research efforts in the study of new EOS models and pressure-induced compounds and materials.

Since no actual experiments are done at 0 K, we have reduced finite-temperature isotherms to 0 K by approximate lattice-dynamics models. There are many sources of error in such a compilation: systematic errors in diamond-cell pressure scales, non-hydrostatic states of compression, approximations in reducing shock compression to isotherms, the Grüneisen model approximation, etc. This tabulation should be regarded as a first attempt to bring a large experimental data set into a common tabular format

for easy use by researchers. We hope that it will stimulate new high-pressure experimental and theoretical work on the elements.

2. THEORETICAL MODELS

2.1 Introduction

We consider three methods for generating 0 K isotherms: 1) Debye-Grüneisen model; 2) reduction of shock data; and 3) density-functional calculations.

2.2 Debye-Grüneisen model

The most common method for generating 0 K isotherms is the Debye-Grüneisen model ⁸. This model assumes that the atomic vibrations are harmonic and that the frequency spectrum can be approximated by a quadratic form.

The Helmholtz free energy for molar volume V and temperature T is given by

$$A(V, T) = E_0(V) + \int_0^{\nu_m} \frac{9N\nu^2}{\nu_m^3} \left[\frac{h\nu}{2} + kT \ln \left(1 - e^{-\frac{h\nu}{kT}} \right) \right] d\nu \quad (1)$$

Here N is the Avogadro number, ν_m is the maximum frequency, and E_0 is the static lattice energy.

The Debye temperature is

$$\Theta_D = \frac{h\nu_m}{k} \quad (2)$$

Then

$$A(V, T) = E_0(V) + \frac{9}{8} Nk\Theta_D + \frac{9NkT}{V_m^3} \int_0^{V_m} v^2 \ln(1 - e^{-\frac{hv}{kT}}) dv \quad (3)$$

The first term is the static-lattice energy, the second is the zero-point energy and the third is the thermal free energy. The pressure is now

$$P = - \left(\frac{\partial A}{\partial V} \right)_T \quad (4)$$

The Debye temperature $\Theta_D(V)$ is assumed to be volume-dependent only, and its logarithmic volume derivative is the Grüneisen function:

$$\gamma(V) = - \frac{d \ln \Theta_D}{d \ln V} \quad (5)$$

We use $\gamma(V)$ as a fitting function to get good agreement with measured thermal expansion data at atmospheric pressure. A useful empirical form for $\gamma(V)$ is

$$\gamma(V) = \frac{1}{2} + \left(\gamma_0 - \frac{1}{2} \right) \left(\frac{V}{V_0} \right)^m \quad (6)$$

where γ_0 is the value of γ at the reference volume V_0 , and m is a free parameter that determines how rapidly γ decreases to the one-component plasma pressure-ionization limit of 0.5.

The thermal pressure P_{th} is then

$$P_{th}(V, T) = 9NkT \frac{\gamma(V)}{V} \left[\ln(1 - e^{-x_m}) - \frac{3}{x_m^3} \int_0^{x_m} x^2 \ln(1 - e^{-x}) dx \right] \quad (7)$$

where $x_m = \Theta_D/T$. Given a smooth fit to the experimental compression isotherm P_{exp} at reference temperature T_0 , we can calculate the zero-Kelvin pressure $P(V, 0)$ as

$$P(V, 0) = P_{exp}(V, T_0) - P_{th}(V, T_0) \quad (8)$$

The zero-point pressure is included in $P(V, 0)$. We set $T_0 = 295$ K unless otherwise specified.

Empirical fits to room-temperature pressure-volume data for low-melting molecular solids give unrealistic extrapolated $P = 0$ volumes, since these states are liquid or gas at room temperature. A realistic alternative is to use a pair-interaction potential to generate a static lattice isotherm, and then adjust the potential parameters to fit the room-temperature experimental data.

We choose the three-parameter exponential-six (“exp-6”) potential ⁹, which has been found useful in modeling high-pressure experiments:

$$\phi(r) = \varepsilon \left\{ \frac{6}{\alpha - 6} \exp \left[\alpha \left(1 - \frac{r}{r_m} \right) \right] - \frac{\alpha}{\alpha - 6} \left(\frac{r_m}{r} \right)^6 \right\} \quad (9)$$

The molar static lattice energy is then

$$E_0(V) = \frac{1}{2} N \sum_i n_i \phi(r_i) \quad (10)$$

where the sum is over all neighbor atom (or molecule) shells in the chosen lattice. For a face-centered cubic lattice, the nearest-neighbor distance r_1 is related to the molar volume by

$$r_1 = 2^{1/6} \left(\frac{V}{N} \right)^{1/3} \quad (11)$$

Further neighbor shell distances r_i are simple multiples of r_1 . For accurate convergence, the sum is taken over the first 40 neighbor shells¹⁰. The static lattice pressure is obtained by numerical differentiation of the energy:

$$P_0(V) = - \frac{dE_0(V)}{dV} \quad (12)$$

We then use the Debye-Grüneisen model to generate the total pressure at the experimental reference temperature T_0 :

$$P(V, T_0) = P_0(V) + \frac{9\gamma(V)}{8V} Nk\Theta_D + 9Nk \frac{\gamma(V)}{V} \left[\frac{3}{x_m^3} \int_0^x x^2 \ln(1 - e^{-x}) - \ln(1 - e^{-x_m}) \right] \quad (13)$$

For a given parameter set (ϵ, r_m, α) , this pressure can be compared with room-temperature compression data and lower-temperature thermal expansion data, and adjustments in the parameters can then be made to achieve good agreement. This pressure function can be reliably extrapolated to 100 GPa. Note that for diatomic molecular solids, such as H_2 or N_2 , the γ function includes librational as well as center-of-mass contributions to the pressure.

2.3 Shock compression data reduction

The second approach is to reduce experimental shock-compression data to the isotherm. We first compute the room-temperature compression adiabat. This can be useful for fitting ramp-compression data that are well-approximated by the adiabat¹¹.

By combining the Rankine-Hugoniot equation that relates the shock pressure, volume, and energy (subscript “H”) with the Grüneisen model, we can solve for the compression adiabat (subscript “ad”):

$$E_H - E_{in} = \frac{1}{2}(P_H + P_{in})(V_{in} - V) \quad (14)$$

Here the initial state is given by $P_{in} = E_{in} = 0$; $V_{in} = V_0$. The relation between thermal pressure and thermal energy at volume V is given by

$$P_H - P_{ad} = \frac{\gamma(V)}{V} (E_H - E_{ad}) \quad (15)$$

Then since $P_{ad} = -dE_{ad}/dV$, we combine Eqs. 14 and 15 to obtain a differential equation for E_{ad} :

$$\frac{dE_{ad}}{dV} + \frac{\gamma(V)}{V} E_{ad} + P_H - \frac{\gamma(V)}{2V} P_H (V_0 - V) = 0 \quad (16)$$

This equation can be solved numerically for E_{ad} and P_{ad} .

The temperature on the adiabat can be computed as follows

$$dS = \left(\frac{\partial S}{\partial V} \right)_T dV + \left(\frac{\partial S}{\partial T} \right)_V dT = 0 \quad (17)$$

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V = \frac{\gamma(V) C_V}{V} \quad (18)$$

$$\left(\frac{\partial S}{\partial T} \right)_V = \frac{C_V}{T}$$

$$\int_{T_0}^T \frac{dT}{T} = - \int_{V_0}^V \frac{\gamma(V)}{V} dV \quad (19)$$

$$T = T_0 \left(\frac{V_0}{V} \right)^{0.5} \exp \left\{ \frac{(\gamma_0 - 0.5)}{m} \left[1 - \left(\frac{V}{V_0} \right)^m \right] \right\} \quad (20)$$

This (V, T) adiabatic state can then be reduced to T = 0 with the Debye-Grüneisen model.

This model works best for stiff materials with no phase transitions below 100 GPa. Uncertainties in the correct form of the Grüneisen function, in the size of mechanical strength corrections, and the accuracy of the measured shock parameters make this approach doubtful for accurate prediction of 0 K compression data. We have used it here to compare with and to confirm our static experimental isotherm computations (see Table 3). For stiff solids with bulk moduli $B > 100$ GPa, the shock Hugoniot is close to the room-temperature isotherm and we can expect accurate agreement between the T = 0 K isotherms computed by the two methods. For lower bulk modulus values $B < 100$ GPa, the shock state at 100 GPa may be liquid, and the Debye-Grüneisen assumption of a harmonic solid lattice will then be incorrect.

2.4 New diamond anvil cell (DAC) measurements

One of us (Cynn) has recently made room-temperature DAC compression measurements on the elements Cr and Rh, which lack accurate high-pressure room-temperature compression data near 100 GPa. In both cases, neon was used as pressure medium and copper was the pressure gauge.

All the x-ray diffraction measurements were done at HP-CAT (16 ID-B) Advanced Photon Source (APS) at Argonne National Laboratory, Illinois. The size of the focused x-ray beam was nominally 6 micron and the wavelength was 0.4066 Angstrom (~ 30 keV). Rh and Cr were measured separately. The Pilatus 1M 2-dimensional x-ray diffraction detector was used for diffraction intensity recording and Fit2D software was used to integrate diffraction intensity. Cr was initially loaded at around 11 GPa and compressed to ~ 55 GPa at ambient temperature. The pressure was released to ~ 17 GPa from the highest pressure in 6 steps. Rh was loaded at ~ 31 GPa and compressed to ~ 83 GPa at ambient temperature. The P-V data were fitted accurately with the Birch-Murnaghan function.

2.5 Density Functional Theory (DFT) Calculations

For the elements Po, At, Rn, Fr, Ra, and Ac, there are no high-pressure experimental data available. We (Söderlind and Landa) have therefore performed density-functional band-structure calculations in two approximations in order to obtain zero-Kelvin isotherms of reasonable accuracy over the range $0 \leq P \leq 100$ GPa.

We do not here attempt to find the lowest-energy crystal structures, but rather to compute reasonably accurate isotherms for comparison with the other elements in the Periodic Table. We assume the following structures over the whole pressure range: Po(simple cubic), At(face-centered cubic), Rn (face-centered cubic), Fr(body-centered cubic), Ra(body-centered cubic), and Ac(face-centered cubic).

The reason we are applying two methods is two-fold. First, for increased confidence that the theoretical approach is robust, and second, the

way the relativistic spin-orbit interaction is implemented in the two methods is different and may influence the calculated pressures.

In both approaches, the calculations are performed within the framework of DFT and the necessary assumption for the unknown electron exchange and correlation functional is that of the generalized gradient approximation (GGA) ¹². Although newer varieties of this approximation have been proposed, the GGA remains the best choice for heavy metals ^{13,14}. Our first implementation is based on the full-potential linear muffin-tin orbitals (FPLMTO) method that has recently been described in detail ¹⁵. In addition to the choice of GGA, we have found that no geometrical approximations (full potential), full relativity including spin-orbit coupling, and a well converged basis set are often needed for good accuracy. Specifically, we associate a set of semi-core states $5s$, $5p$, $5d$ and valence states $6s$, $6p$, $6d$, and $5f$ (Po-Rn) or $6s$, $6p$, $5d$ and valence states $7s$, $7p$, $6d$, and $5f$ (Fr-Ac) to two kinetic energy parameters for a so-called double basis set. Test calculations with a larger basis set including also semi-core $4f$ states did not change the results appreciably. None of these elements (Po-Ac) support formation of magnetic moments.

The second implementation is referred to as EMTO and performed using the Green's-function technique based on the improved screened Korringa-Kohn-Rostoker method, where the one-electron potential is represented by optimized overlapping muffin-tin (OOMT) potential spheres ¹⁶. Within the EMTO formalism, the one-electron states are calculated exactly for the OOMT potentials. As an output of the EMTO calculations, one can determine the self-consistent Green's function of the system and the complete, non-spherically symmetric charge density. Finally, the total energy is calculated using the full charge-density technique ¹⁷. We

treat, as the valence states, the $6s$, $6p$, and $5d$ states for Po and At; $6s$ and $6p$ states for Rn; $6s$, $6p$, and $7s$ states for Fr and Ra; and $6s$, $6p$, $6d$, and $7s$ states for Ac. We treat, as the semi-core states, the $5d$ states for Rn, Fr, Ra, and Ac. In the case of the full relativistic (FR) EMTO formalism, the spin-orbit coupling is included through the four-component Dirac equation ¹⁸.

The first technique (FPLMTO) is generally more accurate than the second (EMTO) because of a somewhat better geometrical approximation of charge densities and potentials. For EMTO, however, full relativity is accomplished by solving the four-spinor Dirac equation ¹⁸ without approximation while the FPLMTO applies a perturbative approach ¹⁹ for the spin-orbit interaction. Fortunately, test calculations showed that the latter approximation is very good for these materials.

Overall, the two DFT techniques produce a pressure behavior with compression that is very similar for all the elements (Po-Ac) and we have therefore chosen to show only the FPLMTO results for clarity.

The zero-point pressure is added to the DFT value by using the Debye temperatures and Grüneisen functions of nearby elements in the Periodic Table. We assume that astatine is a monatomic metal, unlike diatomic iodine.

2.6 Empirical Fitting of Experimental Isotherm Data

The finite-strain model has been a valuable approach to fitting diamond-cell P-V compression data ²⁰. There are many versions of the model, but the most commonly used are the Murnaghan ²⁰, Birch-Murnaghan ²⁰, and Vinet ²¹ forms.

Murnaghan;

$$P(V) = \frac{B_0}{B_0'} \left[\left(\frac{V_0}{V} \right)^{B_0'} - 1 \right] \quad (21)$$

Birch-Murnaghan

$$P(V) = \frac{3B_0}{2} \left[\left(\frac{V_0}{V} \right)^{7/3} - \left(\frac{V_0}{V} \right)^{5/3} \right] \left\{ 1 + \frac{3}{4} (B_0' - 4) \left[\left(\frac{V_0}{V} \right)^{2/3} - 1 \right] \right\} \quad (22)$$

Vinet

$$P(V) = 3B_0 \left[1 - \left(\frac{V}{V_0} \right)^{1/3} \right] \left(\frac{V_0}{V} \right)^{2/3} \exp \left[\frac{3}{2} (B_0' - 1) \left(1 - \left(\frac{V}{V_0} \right)^{1/3} \right) \right] \quad (23)$$

Here B_0 is the bulk modulus and $B_0' = dB/dP$.

Where authors have fitted their data with these functions, we have checked the fits for accuracy, and have refitted data where necessary. Extrapolating these functions can be hazardous, and we have attempted to check any extrapolated numbers for reasonable accuracy. For fitting isotherm segments where extrapolation is not required, we have used polynomials in volume.

2.7 The Grüneisen Function

We use thermal expansion data at $P = 1$ atm to fit the Grüneisen function parameters γ_0 and m . The behavior of γ at high pressures is poorly understood, and in the absence of definite information, we assume the exponent parameter in Eq. (6) is $m = 1$. The same Grüneisen function referenced to V_0 is used for all solid phases over the whole pressure range. In Fig. 1 we show measured and theoretical thermal expansion data for several elements.

The effect of the m parameter on reducing 295 K isotherm data to 0 K can be checked by observing the change in predicted 0 K volumes with changes in the m value. For $m = 0.5$, 1.0, and 2.0, we find that for elements of both high and low compressibility, the effect on the 0 K volume is negligible over the whole pressure range. Evidently the thermal pressure at 295 K is too small to show sensitivity to variations in $\gamma(V)$.

For shock reduction calculations, where temperatures can rise to thousands of Kelvins, the sensitivity to the form of the Grüneisen function is large. It would be very useful to use *ab initio* DFT lattice dynamics to compute γ for the elements over large pressure ranges. This might reveal systematic trends and would allow more accurate modeling of the 0 K isotherms.

3. ELEMENT VOLUMES

Each element is discussed below, with references to the experimental data. Not all elements have been measured up to 100 GPa, and extrapolations are then necessary.

In Table 1 we list the highest pressure recorded for each element, together with the pressure gauge. Note that higher pressures may have been obtained in other work, not used as our chosen standard.

The element volumes are summarized in Table 2. The pressure is given in 1 GPa increments over the range $0 \leq P \leq 100$ GPa. The volumes in m^3/Mmol ($1 \text{ m}^3/\text{Mmol} = 1 \text{ cm}^3/\text{mol}$) are given to three decimal digits to indicate the changes in volume expected at the highest pressures. These extra digits are for comparison purposes only and are not to be considered accurate. We use moles of *atoms* in order to compare the elements with each other.

For many of our calculations, we used the thermal expansion data from the *Thermophysical Properties of Matter* multivolume compilation. The general references for metals ²² and non-metals ²³ are given here, and specific references are referred back to these two volumes.

3.1 Hydrogen

The compression data are from Loubeyre, et al. ²⁴. At low pressures where H is fluid, the Vinet fit of Loubeyre, et al. is not applicable, and we fit low-temperature thermal expansion ²⁵ and lower-pressure compression data ²⁶ with an exp-6 potential model. The two model isotherms are then merged.

3.2 Helium

The compression data are from Loubeyre et al.²⁷ and Dewaele²⁸ and are fitted with a Vinet function. The maximum pressure is 165 GPa. Because room-temperature He melts at 11 GPa, the low pressure Vinet parameters are not applicable. Above this pressure, we use the Debye-Grüneisen model to compute the $T = 0$ K isotherm. The low-temperature, low-pressure data of Driessen, et al.²⁹ can then be fitted accurately with the exp-6 model, and the two model isotherms are merged. At $P = 0$, $T = 0$ K, He is a superfluid with a large atomic volume, $27.579 \text{ m}^3/\text{Mmol}$ ³⁰.

3.3 Lithium

The compression data are from Guillaume, et al.³¹. The data are fitted to a Birch-Murnaghan function. The thermal expansion data are from Touloukian, et al.³².

3.4 Beryllium

The compression data are from Lazicki, et al.³³. The data are fitted to a Vinet function. The thermal expansion data are from Touloukian, et al.³⁴.

3.5 Boron

The compression data are from Sanz, et al.³⁵. The data are fitted to a Vinet function. The thermal expansion data are from Lundström, et al.³⁶ and Touloukian, et al.³⁷.

3.6 Carbon

The graphite and diamond phases of carbon are treated separately. Both phases have been studied in great detail. The compression data for

graphite are from Hanfland, et al.³⁸. The thermal expansion data are from Gustafson³⁹ and Kirby, et al.⁴⁰. The compression data for diamond are from Dewaele, et al.⁴¹ and the thermal expansion data are from Touloukian, et al.⁴². The graphite-diamond transition at 0 K occurs at 1.5 GPa, so there are only two graphite points in the table.

3.7 Nitrogen

The nitrogen isotherm is fitted in two parts with the exp-6 potential. The lower pressure part (to 65 GPa) is from Olijnyk⁴³ and the higher-pressure part is from Gregoryanz, et al.⁴⁴. The thermal expansion data are from Scott⁴⁵.

3.8 Oxygen

The low-pressure data to 10 GPa are from Akahama et al.⁴⁶. The high-pressure data to 100 GPa are from Gorelli et al.⁴⁷. The two regions were fitted with the exp-6 potential. The thermal expansion data are from Roder⁴⁸.

3.9 Fluorine

We have not found any published P-V compression data for solid fluorine. An exp-6 potential used for fitting the experimental melting curve⁴⁹ was used to approximate the isotherm. Comparison with the isotherms of N₂ and O₂ shows a reasonable monotone volume relationship among the three elements over the whole pressure range. The thermal expansion data are from Meyer, et al.⁵⁰.

3.10 Neon

The compression data are from Dewaele, et al. ⁴¹. The low-pressure region is fitted with an exp-6 model, and the high-pressure region is fitted with a Vinet function. The thermal expansion data are from Touloukian, et al. ⁵¹.

3.11 Sodium

The compression data are from Hanfland, et al. ⁵². A combined fit to the bcc and fcc phases is made with a modified Vinet function. The thermal expansion data are from Touloukian, et al. ⁵³ and Anderson and Swenson ⁵⁴.

3.12 Magnesium

The compression data are from Stinton, et al. ⁵⁵. We use two Vinet fits: the hcp phase $0 \leq P \leq 50$ GPa and the bcc phase $50 < P \leq 100$ GPa. The thermal expansion data are from Touloukian, et al. ⁵⁶.

3.13 Aluminum

The compression data are from Dewaele, et al. ⁵⁷. The thermal expansion data are from Touloukian, et al. ⁵⁸.

3.14 Silicon

Silicon has seven distinct high-pressure phases below 100 GPa. There are good isotherm data over this range. For phases I, II, and Imma, we use the data of McMahon, et al. ⁵⁹. For phase V we use the data of Olijnyk, et al. ⁶⁰. For phase VI we use the data of Hanfland, et al. ⁶¹. For phases VII and VIII we use the data of Duclos, et al. ⁶². The thermal expansion data for Si phase I are from Touloukian, et al. ⁶³.

3.15 Phosphorus

Phosphorus has three solid phases below 100 GPa. For phases I and II we use the data of Kikegawa and Iwasaki ⁶⁴. For phase III we use the data of Akahama et al. ⁶⁵. The thermal expansion data are from Riedner, et al. ⁶⁶. These numbers disagree with most x-ray diffraction results. We have therefore scaled Riedner's data to the standard room temperature volume.

3.16 Sulfur

There are four phases of sulfur below 100 GPa. The data for phase I are from Wang, et al. ⁶⁷. The data for phases II and III are from Degtyareva, et al. ⁶⁸. The data for phase IV are from Hejny, et al. ⁶⁹. Thermal expansion data are from Kirby, et al. ⁷⁰.

3.17 Chlorine

Compression data for solid chlorine are from Düsing, et al. ⁷¹. Düsing's data go up to 45 GPa, so we use the exp-6 model to fit potential parameters to the data and extrapolate to 100 GPa. The extrapolated Cl P-V curve shows a reasonable monotonic relation to those of O, Br, and I. The thermal expansion data are from Powell, et al. ⁷².

3.18 Argon

The compression data are from Errandonea, et al. ⁷³. An attempt to fit the whole pressure range with the exp-6 potential model was unsuccessful. An accurate fit joined the Birch-Murnaghan function at high pressures to the exp-6 function at low pressures. The thermal expansion data are from Touloukian, et al. ⁷⁴

3.19 Potassium

Potassium has six phases below 100 GPa. For lower-pressure K-I and K-II data, we use McMahon, et al.⁷⁵. For K-III data we use Lundegaard, et al. up to 55 GPa⁷⁶. For pressures up to 100 GPa, we use the data in Degtyareva⁷⁷. Since the low-pressure Vinet fit⁷⁵ uses an inaccurate room-temperature volume, we combine the low-pressure data of Anderson and Swenson⁵⁴ with the Vinet fit at higher pressures for phases I and II. The thermal expansion data are from Anderson and Swenson⁵⁴.

3.20 Calcium

Calcium has three phases below 100 GPa. For phase I at low pressure, we use the data of Anderson, et al.⁷⁸. For phase I at higher pressure, we use the data of Gu, et al.⁷⁹. For phase II, we use the data of Gu, et al.⁷⁹. For phase III we use the data of Gu, et al.⁷⁹ and Yabuuchi, et al.⁸⁰. The thermal expansion data are from Touloukian, et al.⁸¹.

3.21 Scandium

Scandium has two phases below 100 GPa. The data for phase I are from Grosshans and Holzapfel⁸² and Zhao, et al.⁸³. The data for phase II are from Fujihisa, et al.⁸⁴. The thermal expansion data are from Touloukian, et al.⁸⁵.

3.22 Titanium

The compression data are from Dewaele, et al.⁸⁶. The ω phase is assumed to be the stable phase at $T = 0$ and $P = 0$. The thermal expansion data for the α phase are from Touloukian, et al.⁸⁷.

3.23 Vanadium

The compression data are from Ding, et al.⁸⁸. The thermal expansion data are from Touloukian, et al.⁸⁹.

3.24 Chromium

Our new DAC data go to 56 GPa. The data are fitted with a Birch-Murnaghan function and extrapolated to 100 GPa. This isotherm is very close to that extrapolated from Ming and Manghnani⁹⁰, which was measured only to 10 GPa. The thermal expansion data are from Touloukian, et al.⁹¹. Our fitted Birch-Murnaghan parameters are $V_0 = 7.230 \text{ m}^3/\text{Mmol}$, $B_0 = 193.5 \text{ GPa}$, and $B_0' = 4.934$.

3.25 Manganese

The compression data are from Fujihisa and Takemura⁹². The thermal expansion data are from Touloukian, et al.⁹³.

3.26 Iron

Iron has a phase transition from bcc α to hcp ϵ at 13 GPa. The α compression data are from Dewaele, et al.⁹⁴, and the ϵ compression data are from Dewaele, et al.⁹⁵. The thermal expansion data are from Touloukian, et al.⁹⁶.

3.27 Cobalt

The compression data are from Dewaele, et al.⁹⁴. The thermal expansion data are from Touloukian, et al.⁹⁷.

3.28 Nickel

The compression data are from Dewaele, et al.⁹⁴. The thermal expansion data are from Touloukian, et al.⁹⁸.

3.29 Copper

The compression data are from Dewaele, et al.⁵⁷. The thermal expansion data are from Touloukian, et al.⁹⁹.

3.30 Zinc

The compression data are from Dewaele, et al.⁹⁴. The thermal expansion data are from Touloukian, et al.¹⁰⁰.

3.31 Gallium

Gallium has three phases below 100 GPa. For Ga I at very low pressure we use the data of Lyapin, et al.¹⁰¹. For Ga II and III up to 100 GPa, we use the data of Kenichi, et al.¹⁰². The thermal expansion data are from Donohue¹⁰³.

3.32 Germanium

Germanium has four phases below 100 GPa. For phase I we use the data of Menoni, et al.¹⁰⁴. The higher-pressure phases are combined into a single fit from the data of Vohra, et al.¹⁰⁵ and Chang and Cohen¹⁰⁶. The thermal expansion data are from Touloukian, et al.¹⁰⁷.

3.33 Arsenic

The compression data for arsenic are from Greene, et al.¹⁰⁸. Phase III has a complex host-guest structure¹⁰⁹. The volume of III is difficult to

compute, and the smoothed Greene, et al. data were used as the best approximation. The thermal expansion data are from Touloukian, et al. ¹¹⁰.

3.34 Selenium

Selenium has five phases below 100 GPa. The low-pressure data are from Krüger and Holzapfel ¹¹¹. The high-pressure data are from Degtyareva, et al. ⁶⁸. The thermal expansion data are from Touloukian, et al. ¹¹².

3.35 Bromine

The compression data are from Fuji, et al. ¹¹³. These data are fitted with the exp-6 potential model. The thermal expansion data are from Powell, et al. ⁷².

3.36 Krypton

The compression data are from Jephcoat ¹¹⁴. The thermal expansion data are from Touloukian, et al. ¹¹⁵.

3.37 Rubidium

Rubidium has at least six phases below 100 GPa. For phases I and II we use Takemura and Syassen ¹⁰⁶. For phases III and IV we use Nelmes, et al. ¹¹⁷. For phase V we use Winzenick, et al. ¹¹⁸ and Schwarz, et al. ¹¹⁹. For phase VI we use Schwarz, et al. ¹²⁰ for an absolute volume at 48.1 GPa. We use the data of De Panfilis, et al. ¹²¹ for relative volumes up to 100 GPa, and the absolute volumes are then computed from the 48.1 GPa point. The thermal expansion data are from Anderson and Swenson ⁵⁴.

3.38 Strontium

Strontium has at least five phases below 100 GPa. The low pressure phase I and II data are from Anderson, et al.⁷⁸ and Winzenick, et al.¹²². The data for phases III, IV, and V are from Bovornratnaraks, et al.¹²³ and McMahon, et al.¹²⁴. For pressures above 56 GPa we constructed a smooth curve that is bracketed by the Ca and Ba isotherms. The thermal expansion data are from Touloukian, et al.¹²⁵.

3.39 Yttrium

The compression data are from Grosshans and Holzapfel⁸² up to 60 GPa and Samudrala, et al.¹²⁶ up to 100 GPa. The thermal expansion data are from Touloukian, et al.¹²⁷.

3.40 Zirconium

The compression data for lower pressures are from Zhao, et al.¹²⁸ and for higher pressures from Akahama, et al.¹²⁹. The Akahama data are extrapolated from 68 to 100 GPa with a Birch-Murnaghan fit. The thermal expansion data are from Touloukian, et al.¹³⁰.

3.41 Niobium

The compression data are from Kenichi and Singh¹³¹. The thermal expansion data are from Touloukian, et al.¹³².

3.42 Molybdenum

The compression data are from Dewaele, et al.⁹⁴. The thermal expansion data are from Touloukian, et al.¹³³.

3.43 Technetium

The compression data to 70 GPa are from Mast, et al. ¹³⁴. The thermal expansion data are from Marples and Koch ¹³⁵.

3.44 Ruthenium

The compression data are from Cynn, et al. ¹³⁶. The thermal expansion data are from Touloukian, et al. ¹³⁷.

3.45 Rhodium

Our new DAC data go to 83 GPa. The data are fitted with a Birch-Murnaghan function and extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al. ¹³⁸. The Birch-Murnaghan parameters are $V_0 = 8.273 \text{ m}^3/\text{Mmol}$, $B_0 = 270.0$, $B_0' = 4.756$.

3.46 Palladium

The compression data are from Mao, et al. ¹³⁹. The thermal expansion data are from Touloukian, et al. ¹⁴⁰.

3.47 Silver

The compression data are from Dewaele, et al. ⁹⁴. The thermal expansion data are from Touloukian, et al. ¹⁴¹.

3.48 Cadmium

The compression data are from Kenichi ¹⁴². The thermal expansion data are from Touloukian, et al. ¹⁴³.

3.49 Indium

The compression data are from Takemura and Fujihisa ¹⁴⁴. The two phases I and II are fitted separately with Birch-Murnaghan functions. The thermal expansion data are from Touloukian, et al. ¹⁴⁵.

3.50 Tin

The compression data for the four phases of tin below 100 GPa are from Salamat, et al. ¹⁴⁶. The phases are fitted separately in three parts. The thermal expansion data are from Touloukian, et al. ¹⁴⁷.

3.51 Antimony

The compression data for four phases of antimony up to 65 GPa are from Degtyareva, et al. ¹⁰⁹. The fit to Sb III is extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al. ¹⁴⁸.

3.52 Tellurium

Tellurium has five phases below 100 GPa. For the four lower-pressure phases, we use the data of Parthasarathy and Holzapfel ¹⁴⁹, and for the high-pressure phase V, we use the data of Sugimoto, et al. ¹⁵⁰. The thermal expansion data are from Touloukian, et al. ¹⁵¹.

3.53 Iodine

The compression data from zero to 50 GPa are from Takemura, et al. ¹⁵² and Fuji, et al. ¹⁵³. For pressures above 50 GPa the data are from Reichlin, et al. ¹⁵⁴. The thermal expansion data are from Kirby, et al. ¹⁵⁵.

3.54 Xenon

The compression data at low pressure are from Asaumi ¹⁵⁶, who reduced room-temperature data to 0 K with the Debye-Grüneisen model. Higher pressure data are from Cynn, et al. ¹⁵⁷, who fitted separate fcc and hcp phases. The thermal expansion data are from Touloukian, et al. ¹⁵⁸.

3.55 Cesium

Cesium has six phases below 100 GPa. The low-pressure isotherm to 10 GPa is from Anderson and Swenson ¹⁵⁹, Hall, et al. ¹⁶⁰, and Takemura, et al. ¹⁶¹. The higher-pressure isotherms for phases V and VI are from Takemura, et al. ¹⁶². The thermal expansion data are from Anderson and Swenson ¹⁵⁹.

3.56 Barium

Barium has four phases up to 90 GPa. For the lowest-pressure phases I and II we use the data of Kenichi ¹⁶³. For the very complex incommensurate host-guest phase IV, we use the partial data set of Nelmes, et al. ¹⁶⁴ and Loa, et al. ¹⁶⁵, as well as a smooth interpolation between phases II and V. For phase V we use Kenichi ¹⁶³ to 90 GPa and extrapolate to 100 GPa. The thermal expansion data are from Touloukian, et al. ¹⁶⁶.

3.57 Lanthanum

The compression data to 40 GPa are from Grosshans and Holzapfel ⁸². Extrapolation of the data to 100 GPa is made with a Vinet function. As a check on the extrapolation, a comparison of the Ba, La, and Ce isotherms shows that the La isotherm is bracketed between the Ba and Ce isotherms in a reasonable manner. The thermal expansion data are from Touloukian, et al. ¹⁶⁷.

3.58 Cerium

Cerium has a large volume collapse at low temperatures, and the volumes of the dense low-temperature α phase have been difficult to measure because of the incompleteness of the transition. We start with the room-temperature, high-pressure measurements of the α phase by Zachariasen and Ellinger¹⁶⁸ and extrapolate these data to $P = 0$. This volume together with low-temperature measurements can then be fitted with the Debye-Grüneisen model to obtain a Grüneisen γ value. The compression data are from Staun Olsen, et al.¹⁶⁹ and Vohra, et al.¹⁷⁰, and are fitted with a Vinet function. The low-temperature thermal expansion data are from Donohue¹⁷¹ and Koskenmaki and Gschneidner, Jr.¹⁷².

3.59 Praseodymium

The compression data are from Cunningham, et al.¹⁷³ for pressures below 20 GPa and Chesnut and Vohra¹⁷⁴ for pressures above 20 GPa. The thermal expansion data are from Touloukian, et al.¹⁷⁵.

3.60 Neodymium

The compression data are from Chesnut and Vohra¹⁷⁶. The thermal expansion data are from Touloukian, et al.¹⁷⁷.

3.61 Promethium

The only existing compression data for promethium are from Haire, et al. for $0 \leq P \leq 60$ GPa¹⁷⁸. The pressure-volume curves for Nd and Sm are very close, and the Pm data should lie between them. This is not the case for the Pm experimental data above 40 GPa. Therefore we construct the Pm

isotherm by a weighted average of the $T = 0$ K Nd and Sm isotherms, using the measured $P = 0$ volumes at room temperature to determine the weight.

3.62 Samarium

The compression data for Sm are from Zhao, et al. ¹⁷⁹. The thermal expansion data are from Stankus and Tyagel'skii ¹⁸⁰.

3.63 Europium

The compression data for Eu to 92 GPa are from Bi, et al. ¹⁸¹. These data are fitted and extrapolated to 100 GPa. The thermal expansion data are from Donohue ¹⁸² and Touloukian, et al. ¹⁸³.

3.64 Gadolinium

The compression data are from Samudrala, et al. ¹⁸⁴. The thermal expansion data are from Touloukian, et al. ¹⁸⁵.

3.65 Terbium

The compression data are from Cunningham, et al. ¹⁸⁶. The thermal expansion data are from Touloukian, et al. ¹⁸⁷.

3.66 Dysprosium

The compression data are from Patterson, et al. ¹⁸⁸. The thermal expansion data are from Touloukian, et al. ¹⁸⁹.

3.67 Holmium

The compression data are from Grosshans and Holzapfel ⁸², Cunningham, et al. ¹⁹⁰, and Samudrala and Vohra ¹⁹¹. The thermal expansion data are from Touloukian, et al. ¹⁹².

3.68 Erbium

The compression data are from Samudrala, et al. ¹⁹³. The thermal expansion data are from Touloukian, et al. ¹⁹⁴.

3.69 Thulium

The compression data are from Montgomery, et al. ¹⁹⁵. The thermal expansion data are from Touloukian, et al. ¹⁹⁶.

3.70 Ytterbium

The compression data are from Chesnut and Vohra ¹⁹⁷. Below room temperature and at very low pressure, fcc Yb becomes hcp with a slightly higher volume ¹⁹⁸. The tabulated $T = 0$, $P = 0$ volume is corrected by this factor. The thermal expansion data are from Touloukian, et al. ¹⁹⁹.

3.71 Lutetium

The compression data are from Chesnut and Vohra ²⁰⁰ and are fitted in two parts. The thermal expansion data are from Touloukian, et al. ²⁰¹.

3.72 Hafnium

Compression data to 93 GPa are from Pandey, et al. ²⁰². The isotherm is fitted in three parts and extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al. ²⁰³

3.73 Tantalum

The compression data to 94 GPa are from Dewaele, et al.^{57,204}. The Vinet fit is extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al.²⁰⁵.

3.74 Tungsten

The compression data are from Dewaele, et al.⁵⁷. The thermal expansion data are from Touloukian, et al.²⁰⁶.

3.75 Rhenium

The compression data are from Anzellini, et al.²⁰⁷. The thermal expansion data are from Touloukian, et al.²⁰⁸.

3.76 Osmium

The compression data to 75 GPa are from Occelli, et al.²⁰⁹. The Birch-Murnaghan fit is extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al.²¹⁰.

3.77 Iridium

The compression data to 56 GPa are from Cynn, et al.¹³⁶. The Birch-Murnaghan fit is extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al.²¹¹.

3.78 Platinum

The compression data to 94 GPa are from Dewaele, et al.⁵⁷. The Vinet fit is extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al.²¹²

3.79 Gold

The compression data are from Takemura and Dewaele ²¹³. The thermal expansion data are from Touloukian, et al. ²¹⁴.

3.80 Mercury

The compression isotherm of the low-temperature β phase of mercury is estimated from the work of Swenson ²¹⁵. The higher-pressure γ and δ isotherms are from Takemura, et al. ²¹⁶. The thermal expansion data are from Swenson ²¹⁵.

3.81 Thallium

The compression data to 68 GPa are from Schulte and Holzapfel ²¹⁷. The Birch-Murnaghan fit is extrapolated to 100 GPa. The thermal expansion data are from Touloukian, et al. ²¹⁸.

3.82 Lead

The compression data are from Vohra and Ruoff ²¹⁹. The thermal expansion data are from Touloukian, et al. ²²⁰.

3.83 Bismuth

The low-pressure compression data are from McMahon, et al. ²²¹, and the high-pressure data are from Akahama, et al. ²²². The thermal expansion data are from Touloukian, et al. ²²³.

3.84 Polonium

FPLMTO calculations for the simple cubic structure are fitted to a smooth polynomial function and a zero-point pressure term is added.

3.85 Astatine

FPLMTO calculations for the face-centered cubic structure are fitted to a smooth polynomial function and a zero-point pressure term is added.

3.86 Radon

FPLMTO calculations for the face-centered cubic structure are fitted to a smooth polynomial function and a zero-point pressure term is added.

3.87 Francium

FPLMTO calculations for the body-centered cubic structure are fitted to a smooth polynomial function and a zero-point pressure term is added.

3.88 Radium

FPLMTO calculations for the body-centered cubic structure are fitted to a smooth polynomial function and a zero-point pressure term is added.

3.89 Actinium

FPLMTO calculations for the face-centered cubic structure are fitted to a smooth polynomial function and a zero-point pressure term is added.

3.90 Thorium

The compression data are from Vohra and Holzapfel ²²⁴. The thermal expansion data are from Touloukian, et al. ²²⁵.

4. ERRORS

To estimate the accuracy of our calculated isotherms, we need to consider the following sources of error, in decreasing order of importance:

1. Ruby fluorescence and related pressure scales. The ruby scale was originally calibrated by fitting reduced shock data ¹³⁹, and refinements have been made since.
2. Non-hydrostatic state of the medium surrounding the sample in the DAC.
3. Transcribing P-V points from published graphics.
4. Variations in the stated value of the reference volume.
5. The Debye-Grüneisen model approximation, especially the $\gamma(V)$ function.

Our general approach is to consider the latest compression data, so that systematic errors are better understood and compensated for by the authors. Where there are obvious difficulties in compiling data, such as by extrapolation, we make adjustments based on our best estimate of the highest-pressure volumes. Obviously, pressure standards are continuing to improve, and the present tabulation will need revision.

To estimate probable errors in Table 2, it is worthwhile to compare our data from the table with independently determined element volumes at 100 GPa. The entries in Table 3 are mainly from shock reduction calculations using various approximations ²²⁶⁻²²⁹. Our shock-reduction calculations listed in Table 3 were performed for metals with large bulk modulus, no phase transitions, and accurately determined shock parameters. From this we can conclude that the differences are consistent with average Table 2 errors of a few percent in volume.

5. SYSTEMATICS

The data from Table 2 can be plotted in the form volume vs atomic number for each pressure. This shows clearly the periodicity of the element volumes and the convergence of the volumes to a smooth behavior by 100 GPa. Six plots for pressures of 0, 1, 3, 10, 30, and 100 GPa are given in Figs. 2 - 7.

The sharp peaks at 0 GPa are the alkali metals. The parabolic minima are the transition metals. The rare earths form a straight line broken by the divalent elements Eu and Yb. Under pressure, the highest volumes rapidly decrease and the transition metal region expands as the alkali and alkaline-earth metals take on d-bonding character. The non-metallic elements of the right side of the Periodic Table approach close-packed metallic phases and complex crystal structures become simplified at 100 GPa. At 100 GPa, the volume maxima shift from the alkali metals to the alkaline earths (Mg and Ca) and the rare gases (Kr, Xe, and Rn).

The heavy metals Po – Ac show the same pattern as the lighter elements in the same columns of the Periodic Table, with a sharp peak in the volume for the alkali metal Fr. It appears that the Fr volume does not exceed that of Cs, which has the largest elemental volume at $P = 0$. Table 4 shows the equilibrium atomic volumes at ambient pressure calculated by both FPLMTO and EMTO methods together with available experimental data²³⁰. For Po, Rn, and Ra the FPLMTO (EMTO) method gives slightly underestimated (overestimated) value of the atomic volume in comparison with experimental data. In the case of francium metal FPLMTO calculations give higher atomic volume than EMTO calculations but still smaller than the atomic volume of cesium. In the case of actinium metal both methods

overestimate the experimental data, although results of FPLMTO calculations are in better agreement with experiment.

6. DISCUSSION

6.1 Equation of State (EOS)

The 0 K isotherm can be the basis of wide-range EOS development. By adding models of thermal pressure and energy that smoothly pass from the solid to the liquid and gas, one can construct multiphase EOS models²³¹. These can be used in research in planetary interiors, high-velocity impact phenomena, and laser-driven shock waves.

6.2 Megabar Chemistry

One of the immediate applications of the data in Table 1 is to study the stability of compounds when compared to the pure elements. Under pressure the bonding occurring in the elements and the compound change, and this can destabilize the compound, so that it disproportionates into the pure element constituents or into mixtures of pure elements and other compound stoichiometries. The possibility of high-pressure instability might be called “Megabar chemistry” and opens the prospect of new materials stabilized at high pressures.

For a chemical reaction such as $A + B = AB$, where the elements A and B form the compound AB, the 0 K isotherms of A, B, and AB can be used to examine the stability of the compound under pressure.

At 0 K, the enthalpy (= free energy) of reaction is given by

$$\Delta H(P) = \Delta H(0) + \int_0^P \Delta V(P) dP \quad (24)$$

where $\Delta H = H(AB) - H(A) - H(B)$ and $\Delta V = V(AB) - V(A) - V(B)$.

For stable compounds, $\Delta H(0) < 0$, and under pressure ΔH can become more negative or it can cross the $\Delta H = 0$ value and show that the compound has become unstable to disproportionation into the elements. Alternatively, compounds with $\Delta H(0) > 0$ are unstable at $P = 0$, but can be stabilized with pressure.

Several examples of these trends are sketched in Fig. 8. Strongly bonded ionic solids like MgO increase in stability up to 100 GPa. Less strongly bonded solids like CsI and FeI₂ become less stable under pressure and may disproportionate by 100 GPa. Thermodynamically unstable solids that have been synthesized at high temperature, like Fe₃C, are stabilized by pressure. Because the element volumes become more similar at high pressures (see Fig. 7), the volume change ΔV decreases with pressure. If ΔV , initially negative, changes sign, then the compound becomes less stable and may eventually disproportionate as the pressure increases.

From compendia of binary phase diagrams, we see that most pairs of elements form compounds at $P = 0$ ²³². Under pressure, as ΔV decreases toward zero, we might expect that many of the $P = 0$ compounds will disappear and will be replaced either by new compounds, an insoluble mixture of elements or a solid solution of elements ²³³. Alternatively, elements that are unreactive at $P = 0$ may become reactive and form compounds at high pressures ²³⁴. There is a vast domain of new chemistry yet to be explored at high pressure ²³⁵⁻²³⁷.

6.3 Pressures Beyond 100 GPa

Most of the elements have been compressed to 100 GPa in diamond-anvil cells, but few have been compressed to 200 GPa (see Table 1). A systematic DFT model study of element volumes would be useful for

extending the V- Z plots to pressures beyond 100 GPa. How well do the DFT model calculations agree with static experiments at 100 GPa? How well do they agree with shock reduction calculations at higher pressures? Can adiabatic ramp-compression experiments at temperatures below melting be used to explore these pressure ranges? How rapidly do the volumes converge to simple spherical-atom models such as Thomas-Fermi-Dirac? New two-stage diamond cell experiments to 770 GPa ²³⁸ and ramp-compression experiments to 1200 GPa ²³⁹ may point the way to order-of-magnitude higher isothermal pressures above 100 GPa. There are many worthwhile directions for further research.

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REFERENCES

1. Y. S. Touloukian, (ed.), *Thermophysical Properties of Matter* (IFI/Plenum, New York, 1977) 13 volumes.
2. M. W. Chase, Jr., C. A. Davies, J. R. Downey, Jr., D. J. Frurip, R. A. McDonald, and A. N. Syverud, *JANAF Thermochemical Tables Third Edition*, J. Phys. Chem. Ref. Data **14**, Suppl. 1 (1985).
3. D. E. Gray (ed.), *American Institute of Physics Handbook Third Edition* (McGraw-Hill Book Co., New York, 1972).
4. R. Hultgren, P. D. Desai, D. T. Hawkins, M. Gleiser, K. K. Kelley, and D. D. Wagman (eds.), *Selected Values of the Thermodynamic Properties of the Elements* (American Society for Metals, Metals Park, Ohio, 1973).
5. N. B. Vargaftik, *Handbook of Physical Properties of Liquids and Gases : Pure Substances and Mixtures* Second Edition (Hemisphere, Washington, 1975).
6. S. P. Marsh (ed.), *LASL Shock Hugoniot Data* (University of California Press, Berkeley, 1980).
7. R. F. Trunin (ed.), *Experimental Data on Shock Compression and Adiabatic Expansion of Condensed Matter* (RFNC-VNIIEF, Sarov, 2001)
8. O. L. Anderson, *Equations of State of Solids for Geophysics and Ceramic Science* (Oxford, New York, 1994) chap. 2
9. M. Ross, H. K. Mao, P. M. Bell, and J. A. Xu, J. Chem. Phys. **85**, 1028 (1986).
10. J. O. Hirschfelder, C. F. Curtiss, and R. B. Bird, *Molecular Theory of Gases and Liquids* (Wiley, New York, 1964) p. 1037.

11. J.-P. Davis, J. Appl. Phys. **99**, 103512 (2006).
12. J. Perdew, J. A. Chevary, S. H. Vosko, K. A. Jackson, M. R. Pederson, and D. J. Singh, Phys. Rev. B **46**, 6671 (1992).
13. P. Söderlind, O. Eriksson, B. Johansson, and J. M. Wills, Phys. Rev. B **50**, 7291 (1994)
14. P. Söderlind and A. Gonis, Phys. Rev. B **82**, 033102 (2010).
15. J. M. Wills, O. Eriksson, M. Alouani, and D. L. Price, in: *Electronic Structure and Physical Properties of Solids: The Uses of the LMTO Method*, ed. H. Dreyssé, (Springer-Verlag, Berlin, 2000), pp. 148-167.
16. L. Vitos, *Computational Quantum Mechanics for Materials Engineers: The EMT Method and Application*, (Springer-Verlag, London, 2007).
17. J. Kollar, L. Vitos, and H. L. Skriver, in: *Electronic Structure and Physical Properties of Solids: The Uses of the LMTO Method*, ed. H. Dreyssé, (Springer-Verlag, Berlin, 2000), pp. 85-113.
18. L. V. Pourovskii, A. V. Ruban, L. Vitos, H. Ebert, B. Johansson, and I. A. Abrikosov, Phys. Rev. B **71**, 094415 (2005).
19. O. K. Andersen, Phys. Rev. B **12**, 3060 (1975).
20. O. L. Anderson, *Equations of State of Solids for Geophysics and Ceramic Science* (Oxford, New York, 1994) chap. 6
21. P. Vinet, J. R. Smith, J. Ferrante, and J. H. Rose, Phys. Rev. B **35**, 1945 (1987).
22. Y. S. Touloukian, R. K. Kirby, R. E. Taylor, and P. D. Desai, eds., *Thermophysical Properties of Matter* (IFI/Plenum, New York, 1977) Vol. 12 *Thermal Expansion: Metallic Elements and Alloys*.
23. Y. S. Touloukian, R. K. Kirby, R. E. Taylor, and T. Y. R. Lee, eds., *Thermophysical Properties of Matter* (IFI/Plenum, New York, 1977) Vol. 13 *Thermal Expansion: Nonmetallic Solids*.

24. P. Loubeyre, R. LeToullec, D. Häusermann, M. Hanfland, R. J. Hemley, H. K. Mao, and L. W. Finger, *Nature* **383**, 702 (1996).
25. P. C. Souers, *Hydrogen Properties for Fusion Energy* (University of California Press, Berkeley, 1986) p. 78.
26. G. I. Kerley, *Equations of State for Hydrogen and Deuterium* Sandia Report SAND2003-3613 (2003).
27. P. Loubeyre, R. LeToullec, J. P. Pinceaux, H. K. Mao, J. Hu, and R. J. Hemley, *Phys. Rev. Lett.* **71**, 2272 (1993).
28. A. Dewaele, personal communication (2015).
29. A. Driessen, E. van der Poll, and I. F. Silvera, *Phys. Rev. B* **33**, 3269 (1986).
30. R. K. Crawford in *Rare Gas Solids*, M. L. Klein and J. A. Venables, eds. (Academic Press, London, 1977) vol. 2, p. 675.
31. C. L. Guillaume, E. Gregoryanz, O. Degtyareva, M. I. McMahon, M. Hanfland, S. Evans, M. Guthrie, S. V. Sinogeikin, and H.-K. Mao, *Nature Physics*, **7**, 211 (2011).
32. Ref. 3.0.1, p. 186.
33. A. Lazicki, A. Dewaele, P. Loubeyre, and M. Mezouar, *Phys. Rev. B* **86**, 174118 (2012).
34. Ref. 3.0.1, p. 24.
35. D. N. Sanz, P. Loubeyre, and M. Mezouar, *Phys. Rev. Lett.* **89**, 245501 (2002).
36. T. Lundström, B. Lönnberg, and J. Bauer, *J. Alloys Compounds* **267**, 54 (1998).
37. Ref. 3.0.2, p. 12
38. M. Hanfland, H. Beister, and K. Syassen, *Phys. Rev. B* **39**, 12598 (1989).

39. P. Gustafson, Carbon **24**, 169 (1986).
40. R. K. Kirby, T. A. Hahn, and B. D. Rothrock, in *American Institute of Physics Handbook*, D. E. Gray, ed. (McGraw-Hill, New York, 1972) 3rd edition, p. 4-122.
41. A. Dewaele, F. Datchi, P. Loubeyre, and M. Mezouar, Phys. Rev. B **77**, 094106 (2008).
42. Ref. 3.0.2, p. 19.
43. H. Olijnyk, J. Chem. Phys. **93**, 8968 (1990).
44. E. Gregoryanz, A. F. Goncharov, C. Sanloup, M. Somayazulu, H. Mao, and R. J. Hemley, J. Chem. Phys. **126**, 184505 (2007).
45. T. A. Scott, Phys. Repts. **27C**, 89 (1976).
46. Y. Akahama, H. Kawamura, and O. Shimomura, Phys. Rev. B **64**, 054105 (2001).
47. F. A. Gorelli, L. Ulivi, M. Santoro, M. Hanfland, and S. Desgreniers, High Press. Res. **22**, 13 (2002).
48. H. M. Roder, J. Phys. Chem. Ref. Data **7**, 949 (1978).
49. D. A. Young, C-S. Zha, R. Boehler, J. Yen, M. Nicol, A. S. Zinn, D. Schiferl, S. Kinkead, R. C. Hanson, and D. A. Pinnick, Phys. Rev. B **35**, 5353 (1987).
50. L. Meyer, C. S. Barrett, and S. C. Greer, J. Chem. Phys. **49**, 1902 (1968).
51. Ref. 3.0.2, p. 144.
52. M. Hanfland, I. Loa, and K. Syassen, Phys. Rev. B **65**, 184109 (2002).
53. Ref. 3.0.1, p. 310.
54. M. S. Anderson and C. A. Swenson, Phys. Rev. B **28**, 5395 (1983).
55. G. W. Stinton, S. G. MacLeod, H. Cynn, D. Errandonea, W. J. Evans, J. E. Proctor, Y. Meng, and M. I. McMahon, Phys. Rev. B **90**, 134105 (2014).

56. Ref. 3.0.1, p. 194.
57. A. Dewaele, P. Loubeyre, and M. Mezouar, Phys. Rev. B **70**, 094112 (2004).
58. Ref. 3.0.1, p. 2.
59. M. I. McMahon, R. J. Nelmes, N. G. Wright, and D. R. Allan, Phys. Rev. B **50**, 739 (1994).
60. H. Olijnyk, S. K. Sikka, and W. B. Holzapfel, Phys. Lett. **103A**, 137 (1984).
61. M. Hanfland, U. Schwarz, K. Syassen, and K. Takemura, Phys. Rev. Lett. **82**, 1197 (1999).
62. S. J. Duclos, Y. K. Vohra, and A. L. Ruoff, Phys. Rev. B **41**, 12021 (1990).
63. Ref. 3.0.2, p. 154.
64. T. Kikegawa and H. Iwasaki, Acta. Cryst. **B39**, 158 (1983).
65. Y. Akahama, H. Kawamura, S. Carlson, T. Le Bihan, and D. Häusermann, Phys. Rev. B **61**, 3139 (2000).
66. R. J. Riedner, S. R. Srinivasa, L. Cartz, T. G. Worlton, R. Klinger, and R. Beyerlein, AIP Conf. Proc. **17**, 8 (1974).
67. L. Wang, Y. Zhao, R. Lu, Y. Meng, Y. Fan, H. Luo, Q. Cui, and G. Zou, in *High-Pressure Research in Mineral Physics*, M. H. Manghnani and Y. Syono, eds. (Terra Scientific, Tokyo, 1987) p. 299.
68. O. Degtyareva, E. Gregoryanz, H. K. Mao, and R. J. Hemley, High Press. Res. **25**, 17 (2005).
69. C. Hejny, L. F. Lundegaard, S. Falconi, M. I. McMahon, and M. Hanfland, Phys. Rev. B **71**, 020101 (2005).

70. R. K. Kirby, T. A. Hahn, and B. D. Rothrock, in *American Institute of Physics Handbook*, D. E. Gray, ed. (McGraw-Hill, New York, 1972) 3rd edition, p. 4-130.
71. E.-Fr. Düsing, W. A. Grosshans, and W. B. Holzapfel, *J. de Physique* **45**, C8-203 (1984).
72. B. M. Powell, K. M. Heal, and B. H. Torrie, *Molec. Phys.* **53**, 929 (1984).
73. D. Errandonea, R. Boehler, S. Japel, M. Mezouar, and L. R. Benedetti, *Phys. Rev. B* **73**, 092106 (2006).
74. Ref. 3.0.2, p. 2
75. M. I. McMahon, R. J. Nelmes, U. Schwarz, and K. Syassen, *Phys. Rev. B* **74**, 140102 (2006).
76. L. F. Lundegaard, G. W. Stinton, M. Zelazny, C. L. Guillaume, J. E. Proctor, I. Loa, E. Gregoryanz, R. J. Nelmes, and M. I. McMahon, *Phys. Rev. B* **88**, 054106 (2013).
77. V. F. Degtyareva, *Solid State Sci.* **36**, 62 (2014).
78. M. S. Anderson, C. A. Swenson, and D. T. Peterson, *Phys. Rev. B* **41**, 3329 (1990).
79. Q. F. Gu, G. Krauss, Yu. Grin, and W. Steurer, *Phys. Rev. B* **79**, 134121 (2009).
80. T. Yabuuchi, Y. Nakamoto, K. Shimizu, and T. Kikegawa, *J. Phys. Soc. Japan* **74**, 2391 (2005).
81. Ref. 3.0.1, p. 49.
82. W. A. Grosshans and W. B. Holzapfel, *Phys. Rev. B* **45**, 5171 (1992).
83. Y. C. Zhao, F. Porsch, and W. B. Holzapfel, *Phys. Rev. B* **54**, 9715 (1996).

84. H. Fujihisa, Y. Akahama, H. Kawamura, Y. Gotoh, H. Yamawaki, M. Sakashita, S. Takeya, and K. Honda, Phys. Rev. B **72**, 132103 (2005).
85. Ref. 3.0.1, p. 295.
86. A. Dewaele, V. Stutzmann, J. Bouchet, F. Bottin, F. Occelli, and M. Mezouar, Phys. Rev. B **91**, 134108 (2015).
87. Ref. 3.0.1, p. 346.
88. Y. Ding, R. Ahuja, J. Shu, P. Chow, W. Luo, and H-k. Mao, Phys. Rev. Lett. **98**, 085502 (2007).
89. Ref. 3.0.1, p. 373.
90. L-c. Ming and M. H. Manghnani, J. Appl. Phys. **49**, 208 (1978).
91. Ref. 3.0.1, p. 61.
92. H. Fujihisa and K. Takemura, Phys. Rev. B **52**, 13257 (1995).
93. Ref. 3.0.1, p. 201.
94. A. Dewaele, M. Torrent, P. Loubeyre, and M. Mezouar, Phys. Rev. B **78**, 104102 (2008).
95. A. Dewaele, P. Loubeyre, F. Occelli, M. Mezouar, P. I. Dorogokupets, and M. Torrent, Phys. Rev. Lett. **97**, 215504 (2006).
96. Ref. 3.0.1, p. 157.
97. Ref. 3.0.1, p. 70.
98. Ref. 3.0.1, p.225.
99. Ref. 3.0.1, p. 77.
100. Ref. 3.0.1, p. 391.
101. A. G. Lyapin, E. L. Gromnitskaya, O. F. Yagafarov, O. V. Stal'gorova, and V. V. Brazhkin, J. Exp. Theor. Phys. **107**, 818 (2008).
102. T. Kenichi, K. Kazuaki, and A. Masao, Phys. Rev. B **58**, 2482 (1998).

103. J. Donohue, *The Structures of the Elements* (John Wiley, New York, 1974) p. 236.
104. C. S. Menoni, J. Z. Hu, and I. L. Spain, Phys. Rev. B **34**, 362 (1986).
105. Y. K. Vohra, K. E. Brister, S. Desgreniers, A. L. Ruoff, K. J. Chang, and M. L. Cohen, Phys. Rev. Lett. **56**, 1944 (1986).
106. K. J. Chang and M. L. Cohen, Phys. Rev. B **34**, 8581 (1986).
107. Ref. 3.0.1, p. 116.
108. R. G. Greene, H. Luo, and A. L. Ruoff, Phys. Rev. B **51**, 597 (1995).
109. O. Degtyareva, M. I. McMahon, and R. J. Nelmes, High Press. Res. **24**, 319 (2004).
110. Ref. 3.0.2, p. 7.
111. T. Krüger and W. B. Holzapfel, Phys. Rev. Lett. **69**, 305 (1992).
112. Ref. 3.0.2, p. 149.
113. Y. Fuji, K. Hase, Y. Ohishi, H. Fujihisa, N. Hamaya, K. Takemura, O. Shimomura, T. Kikegawa, Y. Amemiya, and T. Matsushita, Phys. Rev. Lett. **63**, 536 (1989).
114. A. P. Jephcoat, Nature **393**, 355 (1998)
115. Ref. 3.0.2, p. 139.
116. K. Takemura and K. Syassen, Solid State Commun. **44**, 1161 (1982).
117. R. J. Nelmes, M. I. McMahon, J. S. Loveday, and S. Rekhi, Phys. Rev. Lett. **88**, 155503 (2002).
118. M. Winzenick, V. Vijayakumar, and W. B. Holzapfel, Phys. Rev. B **50**, 12381 (1994).
119. U. Schwarz, A. Grzechnik, K. Syassen, I. Loa, and M. Hanfland, Phys. Rev. Lett. **83**, 4085 (1999).
120. U. Schwarz, K. Syassen, A. Grzechnik, and M. Hanfland, Solid State Comm. **112**, 319 (1999).

121. S. De Panfilis, F. Gorelli, M. Santoro, L. Ulivi, E. Gregoryanz, T. Irifune, T. Shinmei, I. Kantor, O. Mathon, and S. Pascarelli, *J. Chem. Phys.* **142**, 214503 (2015).
122. M. Winzenick and W. B. Holzapfel, in *High Pressure Science and Technology*, W. A. Trzeciakowski, ed. (Warsaw, 1995) p. 384.
123. T. Bovornratanaraks, D. R. Allan, S. A. Belmonte, M. I. McMahon, and R. J. Nelmes, *Phys. Rev. B* **73**, 144112 (2006).
124. M. I. McMahon, T. Bovornratanaraks, D. R. Allan, S. A. Belmonte, and R. J. Nelmes, *Phys. Rev. B* **61**, 3135 (2000).
125. Ref. 3.0.1, p. 313.
126. G. K. Samudrala, G. M. Tsoi, and Y. K. Vohra, *J. Phys.: Condens. Matter* **24**, 362201 (2012).
127. Ref. 3.0.1, p. 387.
128. Y. Zhao, J. Zhang, C. Pantea, J. Qian, L. L. Daemen, P. A. Rigg, R. S. Hixson, G. T. Gray III, Y. Yang, L. Wang, Y. Wang, and T. Uchida, *Phys. Rev. B* **71**, 184119 (2005).
129. Y. Akahama, M. Kobayashi, and H. Kawamura, *J. Phys. Soc. Japan* **60**, 3211 (1991).
130. Ref. 3.0.1, p. 400.
131. T. Kenichi and A. K. Singh, *Phys. Rev. B* **73**, 224119 (2006).
132. Ref. 3.0.1, p. 236.
133. Ref. 3.0.1, p. 208.
134. D. S. Mast, E. Kim, E. Siska, F. Poineau, K. R. Czerwinski, B. Lavina, and P. L. Forster (preprint, 2016).
135. J. A. C. Marples and C. C. Koch, *Phys. Lett.* **41A**, 307 (1972).
136. H. Cynn, J. E. Klepeis, C-S. Yoo, and D. A. Young, *Phys. Rev. Lett.* **88**, 135701 (2002).

- 137. Ref. 3.0.1, p. 290.
- 138. Ref. 3.0.1, p. 285.
- 139. H. K. Mao, P. M. Bell, J. W. Shaner, and D. J. Steinberg, *J. Appl. Phys.* **49**, 3276 (1978).
- 140. Ref. 3.0.1, p. 248.
- 141. Ref. 3.0.1, p. 298.
- 142. T. Kenichi, *Phys. Rev. B* **56**, 5170 (1997).
- 143. Ref. 3.0.1, p. 40.
- 144. K. Takemura and H. Fujihisa, *Phys. Rev. B* **47**, 8465 (1993).
- 145. Ref. 3.0.1, p. 143.
- 146. A. Salamat, R. Briggs, P. Bouvier, S. Petitgirard, A. Dewaele, M. E. Cutler, F. Corà, D. Daisenberger, G. Garbarino, and P. F. McMillan, *Phys. Rev. B* **88**, 104104 (2013).
- 147. Ref. 3.0.1, p. 339.
- 148. Ref. 3.0.1, p. 13.
- 149. G. Parthasarathy and W. B. Holzapfel, *Phys. Rev. B* **37**, 8499 (1988).
- 150. T. Sugimoto, Y. Akahama, T. Ichikawa, H. Fujihisa, N. Hirao, and Y. Ohishi, *J. Phys: Conf. Series* **500**, 192018 (2014).
- 151. Ref. 3.0.2, p. 163.
- 152. K. Takemura, S. Minomura, O. Shimomura, Y. Fuji, and J. D. Axe, *Phys. Rev. B* **26**, 998 (1982)
- 153. Y. Fuji, K. Hase, N. Hamaya, Y. Ohishi, A. Onodera, O. Shimomura, and K. Takemura, *Phys. Rev. Lett.* **58**, 796 (1987).
- 154. R. Reichlin, A. K. McMahan, M. Ross, S. Martin, J. Hu, R. J. Hemley, H-k. Mao, and Y. Wu, *Phys. Rev. B* **49**, 3725 (1994).

155. R. K. Kirby, T. A. Hahn, and B. D. Rothrock, in *American Institute of Physics Handbook*, D. E. Gray, ed. (McGraw-Hill, New York, 1972) 3rd edition, p. 4-124.
156. K. Asaumi, Phys. Rev. B **29**, 7026 (1984).
157. H. Cynn, C. S. Yoo, B. Baer, V. Iota-Herbei, A. K. McMahan, M. Nicol, and S. Carlson, Phys. Rev. Lett. **86**, 4552 (2001).
158. Ref. 3.0.2, p. 170.
159. M. S. Anderson and C. A. Swenson, Phys. Rev. B **31**, 668 (1985).
160. H. T. Hall, R. Merrill, and J. D. Barnett, Science **146**, 1297 (1964).
161. K. Takemura, S. Minomura, and O. Shimomura, Phys. Rev. Lett. **49**, 1772 (1982).
162. K. Takemura, N. E. Christensen, D. L. Novikov, K. Syassen, U. Schwarz, and M. Hanfland, Phys. Rev. B **61**, 14399 (2000).
163. T. Kenichi, Phys. Rev. B **50**, 16238 (1994).
164. R. J. Nelmes, D. R. Allan, M. I. McMahon, and S. A. Belmonte, Phys. Rev. Lett. **83**, 4081 (1999).
165. I. Loa, R. J. Nelmes, L. F. Lundegaard, and M. I. McMahon, Nature Mater. **11**, 627 (2012).
166. Ref. 3.0.1, p. 21.
167. Ref. 3.0.1, p. 173.
168. W. H. Zachariasen and F. H. Ellinger, Acta Cryst. **A33**, 155 (1977).
169. J. Staun Olsen, L. Gerward, U. Benedict, and J.-P. Itié, Physica **133B**, 129 (1985).
170. Y. K. Vohra, S. L. Beaver, J. Akella, C. A. Ruddle, and S. T. Weir, J. Appl. Phys. **85**, 2451 (1999).
171. J. Donohue, *The Structures of the Elements* (John Wiley, New York, 1974) p. 94.

172. D. C. Koskenmaki and K. A. Gschneidner, Jr., in *Handbook on the Physics and Chemistry of the Rare Earths*, vol. 1, K. A. Gschneidner, Jr. and L. Eyring, eds. (North-Holland, Amsterdam 1978) p. 337.
173. N. C. Cunningham, N. Velisavljevic, and Y. K. Vohra, Phys. Rev. B **71**, 012108 (2005).
174. G. N. Chesnut and Y. K. Vohra, Phys. Rev. B **62**, 2965 (2000).
175. Ref. 3.0.1, p. 272.
176. G. N. Chesnut and Y. K. Vohra, Phys. Rev. B **61**, R3768 (2000).
177. Ref. 3.0.1, p. 219.
178. R. G. Haire, S. Heathman, and U. Benedict, High Press. Res. **2**, 273 (1990).
179. Y. C. Zhao, F. Porsch, and W. B. Holzapfel, Phys. Rev. B **50**, 6603 (1994).
180. S. V. Stankus and P. V. Tyagel'skii, High Temperature **40**, 660 (2002).
181. W. Bi, Y. Meng, R. S. Kumar, A. L. Cornelius, W. W. Tipton, R. G. Hennig, Y. Zhang, C. Chen, and J. S. Schilling, Phys. Rev. B **83**, 104106 (2011).
182. J. Donohue, *The Structures of the Elements* (John Wiley, New York, 1974) p. 103.
183. Ref. 3.0.1, p. 104.
184. G. K. Samudrala, G. M. Tsoi, S. T. Weir, and Y. K. Vohra, High Press. Res. **34**, 385 (2014).
185. Ref. 3.0.1, p. 107.
186. N. C. Cunningham, W. Qiu, K. M. Hope, H-P. Liermann, and Y. K. Vohra, Phys. Rev. B **76**, 212101 (2007).
187. Ref. 3.0.1, p. 323.

188. R. Patterson, C. K. Saw, and J. Akella, J. Appl. Phys. **95**, 5443 (2004).
189. Ref. 3.0.1, p. 92.
190. N. C. Cunningham, W. Qiu, and Y. K. Vohra, High Press. Res. **26**, 43 (2006).
191. G. K. Samudrala and Y. K. Vohra, in *Handbook on the Physics and Chemistry of Rare Earths*, vol. 43, J.-C. G. Bünzli and V. K. Pecharsky, eds. (North-Holland, Amsterdam, 2013) p. 275.
192. Ref. 3.0.1, p. 138.
193. G. K. Samudrala, S. A. Thomas, J. M. Montgomery, and Y. K. Vohra, J. Phys.: Condens. Matter **23**, 315701 (2011).
194. Ref. 3.0.1, p. 98.
195. J. M. Montgomery, G. K. Samudrala, G. M. Tsoi, and Y. K. Vohra, J. Phys.: Condens. Matter **23**, 155701 (2011).
196. Ref. 3.0.1, p. 336.
197. G. N. Chesnut and Y. K. Vohra, Phys. Rev. Lett. **82**, 1712 (1999).
198. J. Donohue, *The Structures of the Elements* (John Wiley, New York, 1974) p. 120.
199. Ref. 3.0.1, p. 382.
200. G. N. Chesnut and Y. K. Vohra, Phys. Rev. B **57**, 10221 (1998).
201. Ref. 3.0.1, p. 190.
202. K. K. Pandey, J. Gyanchandani, M. Somayazulu, G. K. Dey, S. M. Sharma, and S. K. Sikka, J. Appl. Phys. **115**, 233513 (2014).
203. Ref. 3.0.1, p. 134.
204. A. Dewaele, P. Loubeyre, and M. Mezouar, Phys. Rev. B **69**, 092106 (2004).
205. Ref. 3.0.1, p. 316.
206. Ref. 3.0.1, p. 354.

207. S. Anzellini, A. Dewaele, F. Occelli, P. Loubeyre, and M. Mezouar, J. Appl. Phys. **115**, 043511 (2014).
208. Ref. 3.0.1, p. 280.
209. F. Occelli, D. L. Farber, J. Badro, C. M. Aracne, D. M. Teter, M. Hanfland, B. Canny, and B. Couzinet, Phys. Rev. Lett. **93**, 095502 (2004).
210. Ref. 3.0.1, p. 244.
211. Ref. 3.0.1, p. 153.
212. Ref. 3.0.1, p. 254.
213. K. Takemura and A. Dewaele, Phys. Rev. B **78**, 104119 (2008).
214. Ref. 3.0.1, p. 125.
215. C. A. Swenson, Phys. Rev. **111**, 82 (1958).
216. K. Takemura, S. Nakano, Y. Ohishi, Y. Nakamoto, and H. Fujihisa, Mater. Res. Express **2**, 016502 (2015).
217. O. Schulte and W. B. Holzapfel, Phys. Rev. B **55**, 8122 (1997).
218. Ref. 3.0.1, p. 328.
219. Y. K. Vohra and A. L. Ruoff, Phys. Rev. B **42**, 8651 (1990).
220. Ref. 3.0.1, p. 178.
221. M. I. McMahon, O. Degtyareva, and R. J. Nelmes, Phys. Rev. Lett. **85**, 4896 (2000).
222. Y. Akahama, H. Kawamura, and A. K. Singh, J. Appl. Phys. **92**, 5892 (2002).
223. Ref. 3.0.1, p. 33.
224. Y. K. Vohra and W. B. Holzapfel, High Press. Res. **11**, 223 (1993).
225. Ref. 3.0.1, p. 332.
226. R. G. McQueen, in *High-Velocity Impact Phenomena*, R. Kinslow, ed. (Academic Press, New York, 1970). Chap. 7

227. G. C. Kennedy and R. N. Keeler, in *American Institute of Physics Handbook*, D. E. Gray, ed. (McGraw-Hill, New York, 1972) p. 4-96
228. L. V. Al'tshuler, S. E. Brusnikin, and W. A. Kuz'menkov, *J. Appl. Mech. Tech. Phys.* **28**, 129 (1987).
229. Los Alamos National Laboratory Report LA-UR-92-3407, *SESAME: The Los Alamos National Laboratory Equation of State Database* (1995).
230. D. A. Young, *Phase Diagrams of the Elements*, (University of California Press, Berkeley, 1991) p. 271.
231. D. A. Young and E. M. Corey, *J. Appl. Phys.* **78**, 3748 (1995).
232. T. B. Massalski, ed., *Binary Alloy Phase Diagrams* (ASM International, Materials Park, Ohio, 1990) 2nd edition.
233. H. C. Newton, A. Jayaraman, and G. C. Kennedy, *J. Geophys. Res.* **67**, 2559 (1962).
234. T. Atou, M. Hasegawa, L. J. Parker, and J. V. Badding, *J. Am. Chem. Soc.* **118**, 12104 (1996).
235. P. F. McMillan, *High Press. Res.* **23**, 7 (2003).
236. P. F. McMillan, *High Press. Res.* **24**, 67 (2004).
237. V. V. Brazhkin, *High Press. Res.* **27**, 333 (2007).
238. L. Dubrovinsky, N. Dubrovinskaia, E. Bykova, M. Bykov, V. Prakapenka, C. Prescher, K. Glazyrin, H.-P. Liermann, M. Hanfland, M. Ekholm, Q. Feng, L. V. Pourovskii, M. I. Katsnelson, J. M. Wills, and I. A. Abrikosov, *Nature* **525**, 226 (2015).
239. A. Lazicki, J. R. Rygg, F. Coppari, R. Smith, D. Fratanduono, R. G. Kraus, G. W. Collins, R. Briggs, D. G. Braun, D. C. Swift, and J. H. Eggert, *Phys. Rev. Lett.* **115**, 075502 (2015).

TABLE 1
Maximum diamond-cell pressure (in GPa) and pressure gauge for each element.

Element	P _{max}	gauge	Element	P _{max}	gauge	Element	P _{max}	gauge
H	109	ruby	Ga	150	ruby, Re	Pm	60	ruby
He	165	ruby, W	Ge	125	ruby, Pt	Sm	189	ruby
Li	130	Ta	As	122	Au	Eu	92	Pt
Be	95	ruby	Se	150	ruby	Gd	169	Cu
B	110	ruby	Br	80	ruby	Tb	155	Cu
C	58	ruby, SrB ₄ O ₇ , W	Kr	128	--	Dy	119	Cu
N	170	ruby	Rb	101	Ta	Ho	137	Cu
O	110	--	Sr	56	ruby	Er	155	Cu
F	--	--	Y	177	Cu	Tm	195	Cu
Ne	208	ruby, W	Zr	68	Au	Yb	202	Cu
Na	103	ruby, Ta	Mb	145	ruby	Lu	163	Cu
Mg	211	Cu, Ta	Mo	122	ruby	Hf	93	Cu, Au
Al	153	ruby	Tc	67	Au	Ta	101	ruby
Si	248	ruby, Au	Ru	56	ruby	W	153	ruby
P	280	Re	Rh	83	Cu	Re	144	ruby, He, W
S	101	SrB ₄ O ₇ , He	Pd	77	ruby	Os	75	ruby, NaCl
Cl	45	ruby	Ag	122	ruby	Ir	56	ruby
Ar	114	ruby	Cd	174	ruby, Re	Pt	94	ruby
K	112	ruby	In	93	ruby	Au	123	ruby
Ca	139	ruby, Re	Sn	138	SrB ₄ O ₇ , W	Hg	196	Ta, Pt
Sc	101	ruby	Sb	65	--	Tl	68	ruby
Ti	200	ruby, W	Te	330	Pt	Pb	272	Mo
V	76	Pt	I	276	ruby	Bi	222	ruby, Pt
Cr	56	Cu	Xe	127	ruby	Po	--	--
Mn	190	ruby	Cs	184	ruby, Re	At	--	--
Fe	197	ruby, W	Ba	90	ruby	Rn	--	--
Co	65	ruby	La	40	ruby	Fr	--	--
Ni	156	ruby, He	Ce	208	Cu	Ra	--	--
Cu	153	ruby	Pr	103	Cu	Ac	--	--
Zn	156	ruby, He	Nd	155	Cu	Th	300	ruby, Pt

TABLE 2

The atomic volumes of the elements at $T = 0$ K. The units are P(GPa) and V(m³/Mmol).

Element	H	He	Li	Be	B	C	N	O	F	Ne
P	V	V	V	V	V	V	V	V	V	V
0	11.531	27.579	12.705	4.882	4.626	5.277	13.555	10.380	9.572	13.388
1	5.904	7.383	11.843	4.840	4.605	5.150	10.832	8.763	8.005	9.786
2	5.032	6.354	11.184	4.801	4.583	3.400	9.818	8.060	7.343	8.786
3	4.551	5.784	10.652	4.762	4.562	3.392	9.194	7.607	6.919	8.196
4	4.224	5.396	10.208	4.725	4.541	3.385	8.746	7.274	6.609	7.781
5	3.981	5.104	9.828	4.689	4.521	3.378	8.399	7.012	6.365	7.463
6	3.787	4.873	9.497	4.653	4.500	3.371	8.117	6.796	6.165	7.205
7	3.628	4.682	9.204	4.620	4.480	3.363	7.880	6.614	5.997	7.023
8	3.493	4.520	8.941	4.587	4.461	3.356	7.676	6.160	5.850	6.864
9	3.377	4.379	8.704	4.554	4.441	3.349	7.497	6.021	5.721	6.722
10	3.274	4.256	8.487	4.523	4.422	3.342	7.338	5.898	5.606	6.594
11	3.174	4.147	8.289	4.493	4.403	3.336	7.195	5.787	5.503	6.478
12	3.088	4.048	8.105	4.463	4.385	3.329	7.066	5.686	5.409	6.372
13	3.010	3.958	7.935	4.435	4.366	3.322	6.947	5.594	5.323	6.274
14	2.939	3.877	7.777	4.406	4.348	3.316	6.838	5.510	5.244	6.183
15	2.873	3.801	7.629	4.379	4.330	3.309	6.738	5.431	5.170	6.099
16	2.813	3.732	7.490	4.352	4.313	3.302	6.644	5.357	5.101	6.020
17	2.757	3.667	7.359	4.326	4.295	3.296	6.557	5.289	5.037	5.946
18	2.704	3.607	7.235	4.301	4.278	3.290	6.475	5.225	4.977	5.877
19	2.655	3.550	7.118	4.276	4.261	3.283	6.398	5.165	4.920	5.811
20	2.610	3.497	7.006	4.251	4.244	3.277	6.325	5.107	4.867	5.749
21	2.567	3.447	6.900	4.227	4.227	3.271	6.257	5.054	4.816	5.690
22	2.526	3.400	6.799	4.204	4.211	3.265	6.191	5.003	4.769	5.634
23	2.487	3.355	6.703	4.181	4.195	3.259	6.130	4.954	4.723	5.581
24	2.450	3.313	6.611	4.159	4.179	3.253	6.071	4.907	4.679	5.529
25	2.416	3.272	6.522	4.137	4.163	3.247	6.014	4.863	4.638	5.481
26	2.382	3.234	6.438	4.115	4.147	3.241	5.960	4.821	4.598	5.434
27	2.350	3.197	6.356	4.094	4.132	3.235	5.909	4.780	4.559	5.389
28	2.321	3.161	6.278	4.073	4.116	3.229	5.859	4.741	4.522	5.346
29	2.292	3.128	6.202	4.053	4.101	3.223	5.812	4.704	4.487	5.304
30	2.264	3.095	6.129	4.033	4.086	3.218	5.766	4.668	4.453	5.264
31	2.237	3.064	6.059	4.013	4.072	3.212	5.722	4.633	4.420	5.225
32	2.211	3.034	5.990	3.994	4.057	3.206	5.679	4.599	4.389	5.188
33	2.187	3.005	5.925	3.975	4.042	3.201	5.638	4.566	4.359	5.152
34	2.163	2.977	5.861	3.956	4.028	3.195	5.599	4.535	4.329	5.117
35	2.140	2.950	5.799	3.938	4.014	3.190	5.561	4.505	4.300	5.083
36	2.118	2.924	5.739	3.920	4.000	3.184	5.524	4.476	4.272	5.050
37	2.096	2.899	5.681	3.902	3.986	3.179	5.487	4.447	4.245	5.018
38	2.076	2.875	5.624	3.885	3.972	3.174	5.453	4.420	4.220	4.988
39	2.056	2.851	5.569	3.868	3.959	3.168	5.419	4.393	4.194	4.958
40	2.037	2.828	5.515	3.851	3.945	3.163	5.386	4.367	4.170	4.928
41	2.018	2.806	5.463	3.834	3.932	3.158	5.354	4.341	4.146	4.900
42	2.000	2.784	5.412	3.818	3.919	3.153	5.323	4.317	4.123	4.872
43	1.982	2.764	5.363	3.802	3.906	3.147	5.293	4.293	4.100	4.846
44	1.965	2.743	5.314	3.786	3.893	3.142	5.263	4.269	4.078	4.819
45	1.948	2.723	5.267	3.770	3.880	3.137	5.234	4.247	4.056	4.794
46	1.932	2.704	5.221	3.755	3.867	3.132	5.207	4.225	4.035	4.769
47	1.916	2.685	5.176	3.740	3.855	3.127	5.179	4.203	4.016	4.744
48	1.901	2.667	5.132	3.725	3.842	3.122	5.153	4.182	3.995	4.721
49	1.886	2.649	5.089	3.710	3.830	3.117	5.127	4.162	3.976	4.697

TABLE 2 (continued)

Element	H	He	Li	Be	B	C	N	O	F	Ne
P	V	V	V	V	V	V	V	V	V	V
50	1.871	2.632	5.047	3.696	3.818	3.113	5.101	4.141	3.957	4.675
51	1.857	2.615	5.006	3.681	3.805	3.108	5.077	4.122	3.938	4.652
52	1.843	2.598	4.966	3.667	3.793	3.103	5.053	4.103	3.920	4.631
53	1.829	2.582	4.926	3.653	3.782	3.098	5.029	4.083	3.902	4.609
54	1.817	2.566	4.887	3.639	3.770	3.093	5.005	4.065	3.885	4.589
55	1.804	2.550	4.849	3.626	3.758	3.089	4.982	4.047	3.868	4.568
56	1.791	2.535	4.812	3.612	3.747	3.084	4.960	4.029	3.851	4.548
57	1.778	2.520	4.775	3.599	3.735	3.079	4.939	4.012	3.834	4.529
58	1.766	2.506	4.739	3.586	3.724	3.075	4.917	3.995	3.818	4.509
59	1.754	2.492	4.704	3.573	3.713	3.070	4.896	3.978	3.803	4.490
60	1.743	2.478	4.669	3.560	3.701	3.066	4.875	3.962	3.787	4.472
61	1.732	2.464	4.635	3.548	3.690	3.061	4.856	3.946	3.772	4.454
62	1.720	2.450	4.602	3.535	3.679	3.057	4.835	3.930	3.757	4.436
63	1.710	2.437	4.569	3.523	3.668	3.052	4.816	3.914	3.743	4.418
64	1.699	2.424	4.537	3.511	3.658	3.048	4.797	3.900	3.728	4.401
65	1.688	2.412	4.505	3.499	3.647	3.043	4.785	3.885	3.714	4.384
66	1.679	2.399	4.473	3.487	3.636	3.039	4.761	3.870	3.700	4.368
67	1.669	2.387	4.442	3.475	3.626	3.035	4.737	3.856	3.687	4.351
68	1.659	2.375	4.412	3.464	3.615	3.031	4.714	3.841	3.673	4.335
69	1.649	2.363	4.382	3.452	3.605	3.026	4.691	3.828	3.660	4.320
70	1.639	2.352	4.353	3.441	3.595	3.022	4.668	3.813	3.647	4.304
71	1.630	2.341	4.323	3.430	3.585	3.018	4.646	3.800	3.635	4.289
72	1.621	2.329	4.295	3.419	3.575	3.014	4.624	3.787	3.622	4.274
73	1.612	2.318	4.266	3.408	3.565	3.009	4.603	3.774	3.610	4.259
74	1.603	2.308	4.239	3.397	3.555	3.005	4.582	3.761	3.598	4.244
75	1.595	2.297	4.211	3.386	3.545	3.001	4.561	3.748	3.586	4.230
76	1.586	2.287	4.184	3.376	3.535	2.997	4.540	3.736	3.574	4.216
77	1.577	2.276	4.157	3.365	3.525	2.993	4.519	3.724	3.562	4.202
78	1.569	2.266	4.130	3.355	3.516	2.989	4.500	3.712	3.550	4.188
79	1.561	2.256	4.104	3.344	3.506	2.985	4.480	3.700	3.539	4.175
80	1.553	2.246	4.078	3.334	3.497	2.981	4.461	3.688	3.528	4.161
81	1.546	2.237	4.053	3.324	3.487	2.977	4.442	3.676	3.517	4.148
82	1.538	2.227	4.027	3.314	3.478	2.973	4.423	3.665	3.506	4.135
83	1.530	2.218	4.002	3.304	3.469	2.969	4.404	3.654	3.495	4.122
84	1.522	2.209	3.978	3.294	3.460	2.965	4.386	3.643	3.485	4.110
85	1.515	2.199	3.953	3.285	3.451	2.961	4.369	3.632	3.475	4.097
86	1.508	2.190	3.929	3.275	3.441	2.958	4.351	3.621	3.465	4.085
87	1.500	2.182	3.905	3.266	3.432	2.954	4.333	3.611	3.454	4.073
88	1.494	2.173	3.881	3.256	3.424	2.950	4.316	3.600	3.445	4.061
89	1.486	2.164	3.858	3.247	3.415	2.946	4.299	3.590	3.435	4.049
90	1.480	2.156	3.834	3.238	3.406	2.942	4.282	3.579	3.425	4.038
91	1.473	2.147	3.811	3.229	3.397	2.939	4.266	3.570	3.416	4.026
92	1.466	2.139	3.789	3.220	3.389	2.935	4.250	3.559	3.406	4.015
93	1.460	2.131	3.766	3.211	3.380	2.931	4.234	3.550	3.397	4.003
94	1.454	2.123	3.743	3.202	3.371	2.928	4.218	3.540	3.388	3.992
95	1.447	2.115	3.721	3.193	3.363	2.924	4.202	3.530	3.378	3.981
96	1.440	2.107	3.699	3.184	3.354	2.920	4.186	3.521	3.369	3.971
97	1.435	2.099	3.677	3.176	3.346	2.917	4.171	3.511	3.361	3.960
98	1.429	2.092	3.655	3.167	3.338	2.913	4.156	3.502	3.352	3.949
99	1.422	2.084	3.634	3.159	3.330	2.910	4.141	3.493	3.343	3.939
100	1.417	2.077	3.612	3.150	3.321	2.906	4.126	3.484	3.334	3.929

TABLE 2 (continued)

Element	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
P	V	V	V	V	V	V	V	V	V	V
0	22.692	13.743	9.873	12.047	11.292	14.916	16.516	22.566	43.183	25.962
1	20.366	13.359	9.753	11.929	11.017	14.017	15.242	18.868	35.800	24.700
2	18.821	13.026	9.641	11.817	10.775	12.418	14.444	17.326	32.047	23.673
3	17.674	12.731	9.535	11.710	10.559	12.128	13.861	16.343	29.569	22.810
4	16.768	12.467	9.435	11.608	10.366	11.873	13.402	15.624	27.647	22.032
5	16.023	12.228	9.340	11.509	10.190	11.645	13.024	15.060	26.104	21.323
6	15.394	12.010	9.250	11.415	8.893	11.440	12.701	14.597	24.821	20.688
7	14.851	11.809	9.164	11.324	8.762	11.253	12.422	14.206	23.728	20.115
8	14.375	11.623	9.083	11.236	8.637	11.081	12.175	13.867	22.780	19.593
9	13.952	11.450	9.004	11.152	8.520	10.923	11.954	13.569	21.946	19.114
10	13.572	11.289	8.929	11.070	8.327	10.776	11.755	13.304	21.202	18.671
11	13.229	11.137	8.857	10.992	8.258	10.639	11.572	13.072	20.533	18.259
12	12.916	10.994	8.788	8.391	8.191	10.511	11.405	12.840	19.927	17.875
13	12.628	10.859	8.721	8.360	8.127	10.391	11.251	12.627	19.373	17.515
14	12.363	10.730	8.657	8.270	8.066	10.277	11.107	12.429	18.864	17.176
15	12.117	10.609	8.595	8.208	8.007	10.169	10.973	12.245	18.395	16.856
16	11.888	10.493	8.536	8.026	7.950	10.067	10.847	12.073	17.959	16.553
17	11.674	10.382	8.478	7.973	7.895	9.971	10.729	11.912	17.553	16.265
18	11.473	10.277	8.422	7.921	7.842	9.878	10.617	11.760	17.174	15.991
19	11.284	10.175	8.368	7.873	7.790	9.790	10.512	11.617	15.635	15.730
20	11.106	10.078	8.315	7.826	7.741	9.706	10.412	11.482	15.012	15.068
21	10.938	9.985	8.264	7.781	7.692	9.625	10.316	11.353	14.591	14.787
22	10.778	9.895	8.215	7.737	7.646	9.548	10.226	11.231	14.250	14.540
23	10.626	9.809	8.167	7.695	7.600	9.474	10.139	11.115	13.956	14.317
24	10.482	9.725	8.120	7.655	7.556	9.402	10.056	11.004	13.693	14.112
25	10.344	9.645	8.074	7.615	7.514	9.333	9.976	10.898	13.454	13.921
26	10.213	9.567	8.030	7.577	7.472	9.267	9.900	10.796	13.233	13.742
27	10.087	9.492	7.987	7.540	7.432	9.202	9.826	10.699	13.026	13.573
28	9.966	9.419	7.945	7.504	7.392	9.140	9.755	10.605	12.831	13.412
29	9.851	9.348	7.904	7.469	7.354	9.080	9.688	10.515	12.646	13.258
30	9.740	9.279	7.863	7.435	7.316	9.022	9.622	10.429	12.469	13.110
31	9.633	9.213	7.824	7.401	7.280	8.966	9.559	10.345	12.300	12.969
32	9.530	9.148	7.786	7.369	7.244	8.911	9.497	10.265	12.138	12.832
33	9.431	9.085	7.749	7.337	7.210	8.858	9.438	10.187	11.982	11.929
34	9.335	9.024	7.712	7.306	7.176	8.807	9.380	10.112	11.831	11.791
35	9.243	8.964	7.676	7.275	7.142	8.757	9.324	10.039	11.685	11.665
36	9.154	8.906	7.641	7.246	7.110	8.198	9.270	9.969	11.543	11.550
37	9.067	8.850	7.607	7.216	7.078	8.143	9.217	9.901	11.406	11.442
38	8.984	8.795	7.574	6.923	7.047	8.089	9.166	9.835	11.272	11.341
39	8.903	8.741	7.541	6.897	7.016	8.036	9.117	9.770	11.141	11.245
40	8.824	8.688	7.509	6.870	6.987	7.984	9.068	9.708	11.014	11.154
41	8.748	8.637	7.477	6.842	6.957	7.933	9.021	9.647	10.889	11.067
42	8.674	8.587	7.446	6.814	6.929	7.883	8.975	9.588	10.767	10.983
43	8.602	8.538	7.416	6.785	6.901	7.834	8.931	9.531	10.648	10.902
44	8.532	8.490	7.386	6.572	6.873	7.786	8.886	9.475	10.531	10.825
45	8.464	8.443	7.357	6.543	6.846	7.740	8.844	9.421	10.417	10.749
46	8.398	8.397	7.328	6.514	6.820	7.694	8.803	9.368	10.304	10.676
47	8.333	8.352	7.300	6.486	6.793	7.649	8.762	9.316	10.194	10.606
48	8.270	8.308	7.272	6.458	6.768	7.606	8.722	9.265	10.086	10.537
49	8.209	8.265	7.245	6.431	6.743	7.563	8.683	9.216	9.979	10.470

TABLE 2 (continued)

Element	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
P	V	V	V	V	V	V	V	V	V	V
50	8.149	8.222	7.218	6.404	6.718	7.521	8.645	9.168	9.874	10.404
51	8.091	8.181	7.192	6.378	6.694	7.480	8.608	9.121	9.771	10.340
52	8.034	8.140	7.166	6.352	6.670	7.440	8.572	9.075	9.669	10.278
53	7.979	8.100	7.140	6.326	6.647	7.401	8.536	9.030	9.569	10.216
54	7.924	8.061	7.115	6.301	6.623	7.363	8.501	8.986	9.456	10.156
55	7.871	8.022	7.090	6.277	6.601	7.325	8.466	8.943	9.308	10.098
56	7.820	7.985	7.066	6.252	6.578	7.288	8.433	8.901	9.176	10.040
57	7.769	7.948	7.042	6.228	6.556	7.252	8.400	8.859	9.056	9.984
58	7.719	7.911	7.019	6.205	6.535	7.216	8.367	8.819	8.948	9.928
59	7.671	7.875	6.995	6.182	6.513	7.182	8.336	8.779	8.848	9.873
60	7.623	7.840	6.973	6.159	6.493	7.148	8.305	8.740	8.755	9.820
61	7.576	7.805	6.950	6.136	6.472	7.114	8.274	8.702	8.669	9.767
62	7.531	7.771	6.928	6.114	6.451	7.081	8.244	8.665	8.588	9.715
63	7.486	7.738	6.906	6.092	6.431	7.049	8.214	8.628	8.512	9.663
64	7.442	7.704	6.884	6.071	6.412	7.017	8.185	8.592	8.440	9.613
65	7.399	7.672	6.863	6.049	6.392	6.986	8.157	8.557	8.371	9.563
66	7.357	7.640	6.842	6.029	6.373	6.955	8.129	8.522	8.306	9.514
67	7.316	7.608	6.822	6.008	6.354	6.925	8.101	8.488	8.244	9.465
68	7.275	7.577	6.801	5.988	6.335	6.895	8.074	8.454	8.184	9.417
69	7.235	7.547	6.781	5.967	6.317	6.866	8.047	8.421	8.127	9.370
70	7.196	7.517	6.761	5.948	6.299	6.837	8.021	8.389	8.072	9.323
71	7.158	7.487	6.742	5.928	6.281	6.809	7.995	8.357	8.019	9.277
72	7.120	7.458	6.722	5.909	6.263	6.781	7.970	8.326	7.967	9.232
73	7.083	7.429	6.703	5.890	6.246	6.754	7.944	8.295	7.918	9.187
74	7.046	7.401	6.684	5.871	6.228	6.727	7.920	8.264	7.870	9.142
75	7.010	7.373	6.666	5.852	6.211	6.700	7.896	8.235	7.823	9.098
76	6.975	7.345	6.647	5.834	6.194	6.674	7.872	8.205	7.778	9.055
77	6.940	7.318	6.629	5.816	6.178	6.648	7.848	8.176	7.733	9.012
78	6.906	7.291	6.611	5.798	6.161	6.623	7.824	8.148	7.691	8.969
79	6.873	7.264	6.593	5.753	6.145	6.597	7.801	8.120	7.649	8.927
80	6.840	7.238	6.576	5.738	6.129	6.573	7.779	8.092	7.608	8.885
81	6.807	7.212	6.559	5.722	6.113	6.548	7.756	8.065	7.568	8.843
82	6.775	7.187	6.542	5.707	6.098	6.524	7.734	8.038	7.529	8.802
83	6.744	7.162	6.525	5.692	6.082	6.239	7.713	8.012	7.491	8.762
84	6.713	7.137	6.508	5.677	6.067	6.218	7.691	7.986	7.454	8.722
85	6.682	7.112	6.491	5.662	6.052	6.197	7.670	7.960	7.418	8.682
86	6.652	7.088	6.475	5.647	6.037	6.176	7.649	7.935	7.382	8.642
87	6.622	7.064	6.459	5.633	6.022	6.156	7.628	7.910	7.347	8.603
88	6.593	7.040	6.443	5.619	6.008	6.137	7.607	7.885	7.313	8.564
89	6.564	7.017	6.427	5.604	5.993	6.117	7.588	7.861	7.280	8.526
90	6.536	6.994	6.411	5.590	5.979	6.099	7.567	7.837	7.247	8.487
91	6.508	6.971	6.396	5.577	5.965	6.080	7.548	7.813	7.214	8.449
92	6.480	6.949	6.381	5.563	5.951	6.062	7.529	7.790	7.182	8.412
93	6.453	6.926	6.366	5.549	5.937	6.045	7.510	7.767	7.151	8.375
94	6.426	6.904	6.351	5.536	5.923	6.027	7.490	7.744	7.120	8.338
95	6.400	6.882	6.336	5.523	5.910	6.011	7.471	7.722	7.090	8.301
96	6.373	6.861	6.321	5.509	5.897	5.994	7.453	7.699	7.060	8.264
97	6.348	6.840	6.307	5.496	5.883	5.978	7.435	7.677	7.031	8.228
98	6.322	6.818	6.292	5.483	5.870	5.962	7.417	7.656	7.002	8.192
99	6.297	6.798	6.278	5.471	5.857	5.946	7.399	7.634	6.974	8.157
100	6.272	6.777	6.264	5.458	5.844	5.931	7.381	7.613	6.946	8.121

TABLE 2 (continued)

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
P	V	V	V	V	V	V	V	V	V	V
0	14.959	10.451	8.283	7.202	7.267	7.036	6.638	6.543	7.043	8.972
1	14.722	10.357	8.233	7.167	7.224	6.996	6.604	6.509	6.994	8.846
2	14.498	10.268	8.184	7.132	7.183	6.957	6.570	6.476	6.947	8.730
3	14.286	10.181	8.136	7.098	7.144	6.919	6.538	6.444	6.902	8.623
4	14.085	10.098	8.090	7.065	7.105	6.883	6.507	6.413	6.858	8.524
5	13.893	10.018	8.045	7.033	7.067	6.848	6.476	6.383	6.816	8.431
6	13.711	9.941	8.001	7.002	7.031	6.814	6.446	6.354	6.775	8.344
7	13.537	9.866	7.958	6.971	6.995	6.781	6.417	6.326	6.736	8.262
8	13.371	9.793	7.917	6.941	6.961	6.749	6.388	6.298	6.697	8.185
9	13.212	9.723	7.876	6.912	6.927	6.718	6.361	6.271	6.660	8.112
10	13.059	9.655	7.836	6.884	6.894	6.688	6.334	6.244	6.625	8.042
11	12.912	9.589	7.797	6.856	6.862	6.658	6.307	6.218	6.590	7.976
12	12.770	9.525	7.759	6.829	6.831	6.630	6.281	6.193	6.556	7.912
13	12.634	9.463	7.722	6.803	6.800	6.602	6.256	6.168	6.523	7.851
14	12.503	9.402	7.686	6.777	6.770	6.264	6.231	6.144	6.491	7.793
15	12.376	9.343	7.650	6.752	6.741	6.238	6.207	6.120	6.460	7.737
16	12.253	9.286	7.616	6.727	6.712	6.213	6.183	6.097	6.429	7.684
17	12.134	9.230	7.582	6.702	6.684	6.189	6.159	6.074	6.399	7.632
18	12.019	9.175	7.548	6.679	6.656	6.164	6.136	6.052	6.370	7.582
19	11.908	9.122	7.515	6.655	6.629	6.141	6.114	6.030	6.342	7.534
20	11.800	9.070	7.483	6.632	6.603	6.118	6.092	6.009	6.314	7.487
21	11.695	9.019	7.452	6.610	6.577	6.095	6.070	5.988	6.287	7.442
22	11.593	8.970	7.421	6.588	6.552	6.073	6.049	5.967	6.261	7.398
23	10.906	8.921	7.391	6.566	6.527	6.051	6.028	5.947	6.235	7.356
24	10.817	8.873	7.361	6.544	6.502	6.030	6.007	5.927	6.210	7.315
25	10.727	8.827	7.332	6.524	6.478	6.009	5.987	5.907	6.185	7.275
26	10.634	8.781	7.303	6.503	6.455	5.989	5.967	5.888	6.160	7.236
27	10.541	8.737	7.275	6.483	6.432	5.969	5.948	5.869	6.137	7.199
28	10.447	8.693	7.247	6.463	6.409	5.949	5.929	5.850	6.113	7.162
29	10.353	8.650	7.220	6.443	6.386	5.930	5.910	5.832	6.090	7.126
30	10.260	8.608	7.193	6.424	6.364	5.911	5.891	5.814	6.068	7.091
31	10.167	8.567	7.167	6.405	6.343	5.892	5.873	5.796	6.046	7.057
32	10.077	8.527	7.141	6.386	6.321	5.873	5.855	5.779	6.024	7.024
33	9.988	8.487	7.115	6.368	6.300	5.855	5.837	5.762	6.003	6.992
34	9.902	8.448	7.090	6.350	6.280	5.838	5.820	5.745	5.982	6.960
35	9.818	8.410	7.065	6.332	6.259	5.820	5.802	5.728	5.961	6.929
36	9.737	8.372	7.041	6.315	6.239	5.803	5.785	5.712	5.941	6.899
37	9.659	8.335	7.017	6.297	6.220	5.786	5.769	5.696	5.921	6.869
38	9.584	8.299	6.993	6.280	6.200	5.769	5.752	5.680	5.902	6.840
39	9.511	8.263	6.970	6.263	6.181	5.753	5.736	5.664	5.882	6.812
40	9.441	8.228	6.947	6.247	6.162	5.737	5.720	5.648	5.864	6.784
41	9.373	8.194	6.925	6.230	6.143	5.721	5.704	5.633	5.845	6.757
42	9.308	8.160	6.902	6.214	6.125	5.705	5.688	5.618	5.827	6.730
43	9.245	8.126	6.880	6.198	6.107	5.689	5.673	5.603	5.808	6.704
44	9.184	8.093	6.859	6.183	6.089	5.674	5.658	5.588	5.791	6.678
45	9.126	8.061	6.837	6.167	6.072	5.659	5.643	5.574	5.773	6.653
46	9.069	8.029	6.816	6.152	6.054	5.644	5.628	5.560	5.756	6.628
47	9.014	7.998	6.795	6.137	6.037	5.630	5.613	5.546	5.739	6.604
48	8.961	7.967	6.775	6.122	6.020	5.615	5.599	5.532	5.722	6.580
49	8.909	7.936	6.754	6.107	6.003	5.601	5.584	5.518	5.705	6.557

TABLE 2 (continued)

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
P	V	V	V	V	V	V	V	V	V	V
50	8.859	7.906	6.734	6.093	5.987	5.587	5.570	5.504	5.689	6.533
51	8.811	7.876	6.715	6.079	5.971	5.573	5.556	5.491	5.673	6.511
52	8.764	7.847	6.695	6.064	5.954	5.559	5.542	5.477	5.657	6.489
53	8.718	7.818	6.676	6.050	5.939	5.546	5.529	5.464	5.641	6.467
54	8.673	7.790	6.657	6.037	5.923	5.532	5.515	5.451	5.626	6.445
55	8.630	7.762	6.638	6.023	5.907	5.519	5.502	5.439	5.611	6.424
56	8.587	7.734	6.619	6.009	5.892	5.506	5.489	5.426	5.596	6.403
57	8.546	7.707	6.601	5.996	5.877	5.493	5.476	5.413	5.581	6.382
58	8.506	7.680	6.583	5.983	5.862	5.480	5.463	5.401	5.566	6.362
59	8.466	7.653	6.565	5.970	5.847	5.468	5.450	5.389	5.551	6.342
60	8.428	7.627	6.547	5.957	5.832	5.455	5.438	5.377	5.537	6.322
61	8.390	7.601	6.530	5.944	5.818	5.443	5.425	5.365	5.523	6.303
62	8.353	7.576	6.512	5.932	5.804	5.431	5.413	5.353	5.509	6.284
63	8.317	7.550	6.495	5.919	5.789	5.419	5.401	5.341	5.495	6.265
64	8.282	7.525	6.478	5.907	5.775	5.407	5.389	5.329	5.481	6.247
65	8.248	7.501	6.461	5.895	5.762	5.395	5.377	5.318	5.468	6.228
66	8.214	7.476	6.445	5.882	5.748	5.383	5.365	5.306	5.455	6.210
67	8.180	7.452	6.428	5.871	5.734	5.372	5.353	5.295	5.441	6.193
68	8.148	7.428	6.412	5.859	5.721	5.360	5.341	5.284	5.428	6.175
69	8.116	7.405	6.396	5.847	5.708	5.349	5.330	5.273	5.415	6.158
70	8.084	7.382	6.380	5.835	5.694	5.338	5.319	5.262	5.403	6.141
71	8.053	7.359	6.365	5.824	5.681	5.327	5.307	5.251	5.390	6.124
72	8.023	7.336	6.349	5.813	5.668	5.316	5.296	5.241	5.377	6.107
73	7.993	7.313	6.334	5.801	5.656	5.305	5.285	5.230	5.365	6.091
74	7.964	7.291	6.318	5.790	5.643	5.294	5.274	5.219	5.353	6.074
75	7.935	7.269	6.303	5.779	5.631	5.284	5.263	5.209	5.341	6.058
76	7.906	7.247	6.288	5.768	5.618	5.273	5.253	5.199	5.329	6.042
77	7.878	7.226	6.274	5.757	5.606	5.263	5.242	5.189	5.317	6.027
78	7.851	7.205	6.259	5.747	5.594	5.253	5.232	5.178	5.305	6.011
79	7.823	7.184	6.244	5.736	5.582	5.242	5.221	5.168	5.294	5.996
80	7.797	7.163	6.230	5.726	5.570	5.232	5.211	5.158	5.282	5.981
81	7.770	7.142	6.216	5.715	5.558	5.222	5.201	5.149	5.271	5.966
82	7.744	7.122	6.202	5.705	5.546	5.212	5.190	5.139	5.259	5.951
83	7.718	7.102	6.188	5.695	5.535	5.203	5.180	5.129	5.248	5.937
84	7.693	7.082	6.174	5.685	5.523	5.193	5.170	5.120	5.237	5.922
85	7.668	7.062	6.160	5.674	5.512	5.183	5.160	5.110	5.226	5.908
86	7.643	7.042	6.147	5.665	5.500	5.174	5.151	5.101	5.215	5.894
87	7.619	7.023	6.133	5.655	5.489	5.164	5.141	5.091	5.205	5.880
88	7.595	7.004	6.120	5.645	5.478	5.155	5.131	5.082	5.194	5.866
89	7.571	6.985	6.107	5.635	5.467	5.146	5.122	5.073	5.183	5.852
90	7.548	6.966	6.094	5.626	5.456	5.136	5.112	5.064	5.173	5.839
91	7.525	6.947	6.081	5.616	5.445	5.127	5.103	5.055	5.163	5.825
92	7.502	6.929	6.068	5.607	5.435	5.118	5.093	5.046	5.152	5.812
93	7.479	6.910	6.055	5.597	5.424	5.109	5.084	5.037	5.142	5.799
94	7.457	6.892	6.042	5.588	5.414	5.100	5.075	5.028	5.132	5.786
95	7.435	6.874	6.030	5.579	5.403	5.091	5.066	5.019	5.122	5.773
96	7.413	6.856	6.017	5.570	5.393	5.083	5.057	5.011	5.112	5.761
97	7.391	6.839	6.005	5.561	5.382	5.074	5.048	5.002	5.102	5.748
98	7.370	6.821	5.993	5.552	5.372	5.065	5.039	4.994	5.093	5.736
99	7.348	6.804	5.981	5.543	5.362	5.057	5.030	4.985	5.083	5.723
100	7.327	6.787	5.969	5.534	5.352	5.048	5.022	4.977	5.073	5.711

TABLE 2 (continued)

Element	Ga	Ge	As	Se	Br	Kr	Rb	Sr	Y	Zr
P	V	V	V	V	V	V	V	V	V	V
0	11.656	13.581	12.822	15.977	19.200	27.059	52.616	33.295	19.731	13.962
1	11.470	13.408	12.609	15.146	18.052	22.541	42.410	30.954	19.292	13.816
2	10.489	13.247	12.412	14.607	17.207	20.668	37.078	29.086	18.887	13.678
3	10.339	13.096	12.229	14.205	16.584	19.477	33.579	27.527	18.513	13.546
4	10.200	12.955	12.056	13.884	16.090	18.606	31.009	26.185	18.165	13.420
5	10.071	12.822	11.895	13.616	15.681	17.924	28.996	25.001	17.839	13.300
6	9.950	12.697	11.742	13.386	15.332	17.363	27.351	23.936	17.533	12.914
7	9.836	12.578	11.598	13.185	15.027	16.889	25.968	22.963	17.245	12.804
8	9.729	12.466	11.461	13.006	14.758	16.480	24.778	22.060	16.972	12.698
9	9.627	12.359	11.331	12.844	14.456	16.120	23.737	21.209	16.714	12.596
10	9.531	12.257	11.207	12.698	14.191	15.799	22.813	20.395	16.468	12.498
11	9.440	9.846	11.088	12.563	13.949	15.509	21.984	19.510	16.233	12.403
12	9.353	9.772	10.975	12.439	13.728	15.247	21.232	18.913	16.010	12.312
13	9.269	9.700	10.866	12.324	13.524	15.006	20.546	18.452	15.795	12.224
14	9.190	9.630	10.762	12.217	13.335	14.784	18.744	18.071	15.590	12.138
15	9.114	9.561	10.662	11.432	13.159	14.579	18.276	17.742	15.392	12.055
16	9.041	9.495	10.566	11.318	12.995	14.387	17.909	17.452	15.202	11.975
17	8.971	9.431	10.473	11.205	12.839	14.208	17.014	17.189	15.019	11.897
18	8.903	9.369	10.383	11.094	12.694	14.041	16.659	16.950	14.842	11.822
19	8.838	9.309	10.297	10.985	12.556	13.882	16.349	16.728	14.672	11.748
20	8.775	9.250	10.213	10.878	12.426	13.733	15.594	16.522	14.506	11.677
21	8.715	9.194	10.132	10.775	12.302	13.592	15.196	16.329	14.346	11.607
22	8.656	9.139	10.054	10.674	12.184	13.457	14.879	16.146	14.191	11.540
23	8.599	9.085	9.978	10.577	12.072	13.330	14.609	15.974	14.040	11.474
24	8.544	9.034	9.904	10.484	11.963	13.208	14.368	15.810	13.894	11.409
25	8.491	8.983	9.833	10.394	11.861	13.091	14.149	15.006	13.752	11.347
26	8.439	8.934	9.763	10.307	11.762	12.980	13.948	14.815	13.613	11.285
27	8.389	8.886	9.696	10.224	11.667	12.873	13.759	14.629	13.478	11.226
28	8.341	8.840	9.630	10.143	11.576	12.771	13.582	14.447	13.346	11.167
29	8.293	8.794	9.566	10.065	11.488	12.672	13.414	14.270	13.218	11.110
30	8.247	8.750	9.503	9.990	11.403	12.577	13.254	14.097	13.092	11.054
31	8.202	8.707	9.443	9.918	11.322	12.486	13.101	13.928	12.970	11.000
32	8.158	8.665	9.383	9.848	11.242	12.398	12.954	13.762	12.850	10.946
33	8.116	8.624	9.325	9.781	11.166	12.312	12.813	13.600	12.732	10.687
34	8.074	8.584	9.269	9.716	11.092	12.230	12.676	13.441	12.617	10.635
35	8.034	8.545	9.214	9.652	11.020	12.150	12.544	13.284	12.505	10.585
36	7.994	8.506	9.160	9.591	10.950	12.073	12.416	13.131	12.394	10.535
37	7.955	8.469	9.107	9.532	10.883	11.998	12.291	12.980	12.286	10.486
38	7.917	8.432	9.056	9.474	10.818	11.926	12.170	12.832	12.179	10.438
39	7.880	8.396	9.005	9.418	10.754	11.855	12.052	12.686	12.075	10.390
40	7.844	8.360	8.956	9.363	10.692	11.786	11.937	12.542	11.972	10.344
41	7.809	8.326	8.908	9.310	10.632	11.720	11.825	12.401	11.871	10.298
42	7.774	8.292	8.860	9.258	10.572	11.655	11.714	12.261	11.771	10.253
43	7.740	8.258	8.814	9.208	10.515	11.592	11.607	12.124	11.673	10.209
44	7.707	8.226	8.768	9.159	10.460	11.530	11.501	11.989	11.577	10.165
45	7.675	8.194	8.724	9.111	10.405	11.470	11.398	11.855	11.482	10.123
46	7.643	8.162	8.680	9.064	10.352	11.412	11.296	11.723	11.388	10.080
47	7.611	8.131	8.637	9.018	10.300	11.355	11.197	11.593	11.295	10.039
48	7.581	8.100	8.595	8.973	10.249	11.299	11.099	11.256	11.204	9.998
49	7.550	8.070	8.554	8.930	10.200	11.245	10.785	11.159	11.113	9.958

TABLE 2 (continued)

Element	Ga	Ge	As	Se	Br	Kr	Rb	Sr	Y	Zr
P	V	V	V	V	V	V	V	V	V	V
50	7.521	8.041	8.513	8.887	10.151	11.192	10.741	11.074	11.024	9.918
51	7.492	8.012	8.473	8.845	10.104	11.140	10.697	10.997	10.936	9.879
52	7.463	7.983	8.434	8.803	10.057	11.089	10.654	10.926	10.848	9.841
53	7.435	7.955	8.396	8.763	10.011	11.039	10.612	10.859	10.761	9.803
54	7.408	7.927	8.358	8.723	9.967	10.990	10.571	10.796	10.675	9.765
55	7.381	7.899	8.321	8.685	9.924	10.943	10.531	10.737	10.590	9.729
56	7.354	7.872	8.284	8.646	9.881	10.896	10.491	10.679	10.506	9.508
57	7.328	7.846	8.248	8.609	9.839	10.850	10.452	10.625	10.422	9.475
58	7.302	7.820	8.213	8.572	9.797	10.805	10.414	10.572	10.338	9.443
59	7.277	7.794	8.178	8.536	9.758	10.761	10.376	10.520	10.255	9.412
60	7.252	7.768	8.144	8.500	9.717	10.718	10.339	10.471	10.110	9.381
61	7.227	7.743	8.110	8.465	9.679	10.676	10.303	10.422	10.072	9.351
62	7.203	7.718	8.077	8.431	9.640	10.634	10.266	10.375	10.035	9.320
63	7.179	7.693	8.044	8.397	9.603	10.593	10.231	10.329	9.998	9.291
64	7.156	7.669	8.011	8.363	9.566	10.553	10.196	10.284	9.962	9.261
65	7.133	7.645	7.980	8.330	9.530	10.514	10.161	10.240	9.926	9.232
66	7.110	7.622	7.948	8.298	9.495	10.475	10.127	10.197	9.891	9.204
67	7.088	7.598	7.918	8.266	9.460	10.437	10.094	10.155	9.857	9.176
68	7.066	7.575	7.887	8.235	9.425	10.399	10.060	10.114	9.822	9.148
69	7.044	7.552	7.857	8.204	9.391	10.363	10.028	10.073	9.789	9.120
70	7.022	7.530	7.827	8.173	9.358	10.327	9.995	10.032	9.756	9.093
71	7.001	7.507	7.798	8.143	9.325	10.291	9.963	9.993	9.723	9.066
72	6.980	7.485	7.769	8.113	9.293	10.256	9.931	9.954	9.690	9.040
73	6.960	7.463	7.741	8.083	9.261	10.221	9.900	9.915	9.659	9.013
74	6.939	7.442	7.713	8.054	9.229	10.188	9.869	9.877	9.627	8.987
75	6.919	7.420	7.685	8.026	9.199	10.154	9.838	9.840	9.596	8.962
76	6.900	7.399	7.658	7.997	9.168	10.121	9.808	9.803	9.565	8.937
77	6.880	7.378	7.631	7.969	9.139	10.089	9.778	9.766	9.535	8.912
78	6.861	7.357	7.605	7.942	9.109	10.057	9.748	9.730	9.505	8.887
79	6.842	7.337	7.578	7.914	9.080	10.025	9.719	9.694	9.476	8.862
80	6.823	7.317	7.552	7.887	9.051	9.994	9.690	9.658	9.447	8.838
81	6.804	7.296	7.527	7.861	9.023	9.964	9.661	9.623	9.418	8.814
82	6.786	7.277	7.501	7.834	8.995	9.934	9.632	9.588	9.390	8.790
83	6.768	7.257	7.476	7.808	8.967	9.904	9.604	9.553	9.362	8.767
84	6.750	7.237	7.452	7.782	8.941	9.875	9.576	9.519	9.334	8.744
85	6.732	7.218	7.427	7.757	8.914	9.846	9.548	9.485	9.306	8.721
86	6.715	7.199	7.403	7.732	8.887	9.817	9.520	9.451	9.279	8.698
87	6.698	7.180	7.379	7.707	8.861	9.789	9.493	9.417	9.253	8.676
88	6.681	7.161	7.356	7.682	8.835	9.761	9.465	9.384	9.226	8.654
89	6.664	7.142	7.333	7.657	8.810	9.734	9.438	9.351	9.200	8.632
90	6.647	7.124	7.310	7.633	8.785	9.707	9.412	9.318	9.174	8.610
91	6.631	7.105	7.287	7.609	8.760	9.680	9.385	9.285	9.148	8.588
92	6.614	7.087	7.264	7.585	8.736	9.654	9.359	9.253	9.123	8.567
93	6.598	7.069	7.242	7.562	8.711	9.628	9.333	9.220	9.098	8.546
94	6.582	7.051	7.220	7.538	8.688	9.602	9.307	9.188	9.073	8.525
95	6.567	7.033	7.198	7.515	8.664	9.576	9.281	9.156	9.049	8.504
96	6.551	7.016	7.177	7.492	8.640	9.551	9.256	9.125	9.025	8.484
97	6.536	6.998	7.155	7.470	8.617	9.526	9.230	9.093	9.001	8.463
98	6.520	6.981	7.134	7.447	8.595	9.502	9.205	9.061	8.977	8.443
99	6.505	6.964	7.113	7.425	8.572	9.478	9.180	9.030	8.709	8.423
100	6.490	6.947	7.093	7.403	8.550	9.454	9.155	8.999	8.687	8.404

TABLE 2 (continued)

Element	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn
P	V	V	V	V	V	V	V	V	V	V
0	10.779	9.344	8.589	8.144	8.235	8.794	10.125	12.710	15.416	16.072
1	10.717	9.309	8.560	8.121	8.206	8.749	10.036	12.467	15.101	15.809
2	10.656	9.275	8.532	8.099	8.177	8.706	9.951	12.253	14.819	15.567
3	10.597	9.241	8.505	8.076	8.149	8.664	9.870	12.062	14.564	15.344
4	10.539	9.208	8.478	8.054	8.122	8.623	9.794	11.889	14.330	15.136
5	10.483	9.176	8.452	8.033	8.095	8.583	9.722	11.732	14.115	14.943
6	10.428	9.144	8.426	8.011	8.068	8.544	9.652	11.587	13.916	14.761
7	10.374	9.112	8.401	7.990	8.042	8.507	9.586	11.453	13.730	14.590
8	10.321	9.081	8.376	7.969	8.016	8.470	9.523	11.328	13.557	14.429
9	10.269	9.051	8.352	7.948	7.991	8.434	9.462	11.212	13.395	14.277
10	10.219	9.021	8.328	7.927	7.966	8.399	9.403	11.102	13.242	14.132
11	10.169	8.991	8.305	7.907	7.942	8.366	9.347	10.999	13.097	13.716
12	10.120	8.962	8.282	7.887	7.918	8.332	9.293	10.901	12.960	13.590
13	10.073	8.934	8.259	7.867	7.894	8.300	9.240	10.809	12.830	13.468
14	10.026	8.906	8.237	7.847	7.871	8.268	9.189	10.721	12.706	13.352
15	9.980	8.878	8.215	7.827	7.848	8.237	9.140	10.637	12.588	13.240
16	9.935	8.851	8.194	7.808	7.826	8.207	9.093	10.557	12.475	13.132
17	9.891	8.824	8.173	7.789	7.804	8.177	9.047	10.481	12.367	13.027
18	9.848	8.797	8.152	7.770	7.782	8.148	9.002	10.407	12.264	12.927
19	9.805	8.771	8.132	7.751	7.761	8.120	8.958	10.337	12.164	12.830
20	9.763	8.746	8.112	7.732	7.740	8.092	8.916	10.269	12.069	12.735
21	9.722	8.720	8.092	7.714	7.719	8.065	8.875	10.204	11.977	12.644
22	9.682	8.695	8.072	7.696	7.698	8.038	8.835	10.141	11.888	12.556
23	9.642	8.670	8.053	7.678	7.678	8.012	8.796	10.080	11.802	12.470
24	9.603	8.646	8.034	7.660	7.658	7.986	8.757	10.022	11.719	12.387
25	9.564	8.622	8.016	7.642	7.638	7.961	8.720	9.965	11.639	12.306
26	9.526	8.598	7.997	7.624	7.619	7.936	8.684	9.910	11.562	12.228
27	9.489	8.575	7.979	7.607	7.600	7.911	8.649	9.857	11.487	12.151
28	9.452	8.551	7.961	7.590	7.581	7.887	8.614	9.805	11.414	12.077
29	9.416	8.528	7.944	7.573	7.562	7.864	8.580	9.755	11.344	12.004
30	9.381	8.506	7.926	7.556	7.544	7.841	8.547	9.706	11.275	11.934
31	9.345	8.483	7.909	7.539	7.526	7.818	8.515	9.659	11.208	11.865
32	9.311	8.461	7.892	7.522	7.508	7.795	8.483	9.613	11.143	11.797
33	9.277	8.440	7.876	7.506	7.490	7.773	8.452	9.568	11.080	11.731
34	9.243	8.418	7.859	7.489	7.473	7.752	8.421	9.524	11.019	11.675
35	9.210	8.397	7.843	7.473	7.455	7.730	8.391	9.481	10.959	11.621
36	9.177	8.376	7.827	7.457	7.438	7.709	8.362	9.440	10.901	11.568
37	9.145	8.355	7.811	7.441	7.422	7.688	8.333	9.399	10.844	11.516
38	9.113	8.334	7.795	7.425	7.405	7.668	8.305	9.359	10.788	11.466
39	9.082	8.314	7.780	7.410	7.388	7.648	8.277	9.321	10.734	11.417
40	9.051	8.294	7.764	7.394	7.372	7.628	8.250	9.283	10.681	11.369
41	9.021	8.274	7.749	7.379	7.356	7.609	8.223	9.246	10.629	11.323
42	8.990	8.254	7.734	7.364	7.340	7.589	8.197	9.209	10.579	11.277
43	8.961	8.234	7.720	7.348	7.325	7.570	8.171	9.174	10.529	11.232
44	8.931	8.215	7.705	7.333	7.309	7.552	8.146	9.139	10.481	11.188
45	8.902	8.196	7.690	7.318	7.294	7.533	8.121	9.105	10.433	11.145
46	8.874	8.177	7.676	7.304	7.278	7.515	8.096	9.072	10.387	11.103
47	8.845	8.158	7.662	7.289	7.263	7.497	8.072	9.039	10.341	11.061
48	8.817	8.140	7.648	7.274	7.249	7.479	8.049	9.007	10.296	11.020
49	8.790	8.122	7.634	7.260	7.234	7.462	8.025	8.975	10.253	10.980

TABLE 2 (continued)

Element	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn
P	V	V	V	V	V	V	V	V	V	V
50	8.762	8.103	7.620	7.246	7.219	7.444	8.002	8.945	10.217	10.940
51	8.735	8.085	7.607	7.231	7.205	7.427	7.979	8.914	10.181	10.901
52	8.709	8.068	7.593	7.217	7.191	7.410	7.957	8.885	10.146	10.863
53	8.682	8.050	7.580	7.203	7.176	7.394	7.935	8.855	10.111	10.825
54	8.656	8.033	7.567	7.189	7.162	7.377	7.913	8.827	10.077	10.787
55	8.631	8.015	7.554	7.176	7.149	7.361	7.892	8.798	10.044	10.751
56	8.605	7.998	7.541	7.162	7.135	7.345	7.871	8.771	10.011	10.714
57	8.580	7.981	7.528	7.148	7.121	7.329	7.850	8.743	9.979	10.678
58	8.555	7.964	7.516	7.135	7.108	7.313	7.829	8.716	9.947	10.643
59	8.530	7.948	7.503	7.121	7.095	7.298	7.809	8.690	9.916	10.608
60	8.506	7.931	7.491	7.108	7.081	7.282	7.789	8.664	9.885	10.573
61	8.482	7.915	7.479	7.095	7.068	7.267	7.769	8.639	9.855	10.539
62	8.458	7.899	7.466	7.082	7.055	7.252	7.750	8.613	9.825	10.505
63	8.434	7.882	7.454	7.069	7.043	7.238	7.731	8.589	9.796	10.472
64	8.411	7.867	7.442	7.056	7.030	7.223	7.712	8.564	9.767	10.439
65	8.388	7.851	7.431	7.043	7.017	7.208	7.693	8.540	9.739	10.406
66	8.365	7.835	7.419	7.030	7.005	7.194	7.675	8.516	9.711	10.374
67	8.342	7.820	7.407	7.017	6.993	7.180	7.656	8.493	9.684	10.342
68	8.319	7.804	7.396	7.005	6.980	7.166	7.638	8.470	9.657	10.310
69	8.297	7.789	7.384	6.992	6.968	7.152	7.621	8.447	9.630	10.279
70	8.275	7.774	7.373	6.980	6.956	7.138	7.603	8.425	9.603	10.248
71	8.253	7.759	7.362	6.968	6.944	7.125	7.586	8.403	9.577	10.217
72	8.232	7.744	7.351	6.955	6.932	7.111	7.568	8.381	9.552	10.186
73	8.210	7.729	7.340	6.943	6.921	7.098	7.551	8.360	9.526	10.156
74	8.189	7.714	7.329	6.931	6.909	7.085	7.534	8.339	9.502	10.126
75	8.168	7.700	7.318	6.919	6.898	7.071	7.518	8.318	9.477	10.097
76	8.147	7.685	7.307	6.907	6.886	7.059	7.501	8.297	9.453	10.067
77	8.127	7.671	7.297	6.895	6.875	7.046	7.485	8.277	9.429	10.038
78	8.106	7.657	7.286	6.884	6.864	7.033	7.469	8.257	9.405	10.009
79	8.086	7.643	7.275	6.872	6.853	7.020	7.453	8.237	9.382	9.980
80	8.066	7.629	7.265	6.860	6.841	7.008	7.437	8.217	9.359	9.952
81	8.046	7.615	7.255	6.849	6.831	6.996	7.422	8.198	9.336	9.924
82	8.026	7.601	7.245	6.837	6.820	6.983	7.406	8.179	9.313	9.896
83	8.007	7.588	7.234	6.826	6.809	6.971	7.391	8.160	9.291	9.868
84	7.987	7.574	7.224	6.814	6.798	6.959	7.376	8.141	9.269	9.840
85	7.968	7.561	7.214	6.803	6.788	6.947	7.361	8.123	9.247	9.813
86	7.949	7.547	7.204	6.792	6.777	6.936	7.346	8.105	9.226	9.786
87	7.930	7.534	7.195	6.781	6.767	6.924	7.332	8.087	9.205	9.759
88	7.912	7.521	7.185	6.770	6.756	6.912	7.317	8.069	9.184	9.732
89	7.893	7.508	7.175	6.759	6.746	6.901	7.303	8.051	9.163	9.705
90	7.875	7.495	7.165	6.748	6.736	6.890	7.289	8.034	9.143	9.679
91	7.856	7.482	7.156	6.737	6.726	6.878	7.275	8.017	9.122	9.653
92	7.838	7.469	7.146	6.726	6.716	6.867	7.261	8.000	9.102	9.627
93	7.820	7.457	7.137	6.716	6.706	6.856	7.247	7.983	9.083	9.601
94	7.803	7.444	7.128	6.705	6.696	6.845	7.233	7.966	9.063	9.575
95	7.785	7.432	7.118	6.694	6.686	6.834	7.220	7.950	9.044	9.549
96	7.767	7.419	7.109	6.684	6.676	6.823	7.206	7.933	9.025	9.524
97	7.750	7.407	7.100	6.673	6.666	6.813	7.193	7.917	9.006	9.499
98	7.733	7.395	7.091	6.663	6.657	6.802	7.180	7.901	8.987	9.474
99	7.716	7.382	7.082	6.652	6.647	6.792	7.167	7.885	8.968	9.449
100	7.699	7.370	7.073	6.642	6.638	6.781	7.154	7.870	8.950	9.424

TABLE 2 (continued)

Element	Sb	Te	I	Xe	Cs	Ba	La	Ce	Pr	Nd
P	V	V	V	V	V	V	V	V	V	V
0	18.037	20.196	24.444	34.739	66.383	37.586	22.491	16.911	20.719	20.460
1	17.602	19.329	22.951	30.389	49.806	33.854	21.597	16.308	19.833	19.723
2	17.224	18.729	21.869	28.032	42.874	31.280	20.870	15.827	19.068	19.083
3	16.888	18.262	21.023	26.421	35.877	29.326	20.256	15.426	18.402	18.520
4	16.587	17.878	20.329	25.206	31.663	27.759	19.725	15.082	17.813	18.019
5	16.313	16.877	19.744	24.234	24.691	26.456	19.258	14.782	17.286	17.569
6	16.063	16.662	19.238	23.468	22.847	24.204	18.841	14.514	16.809	17.160
7	15.833	16.193	18.792	22.775	21.820	22.671	18.464	14.273	16.372	16.787
8	15.620	15.912	18.395	22.172	21.081	21.534	18.120	14.055	15.969	16.444
9	14.564	15.701	18.038	21.639	20.491	20.649	17.805	13.854	15.594	16.127
10	14.430	15.526	17.715	21.162	19.993	19.923	17.513	13.669	15.243	15.833
11	14.302	15.121	17.418	20.732	18.740	19.305	17.243	13.497	14.914	15.559
12	14.178	14.950	17.144	20.341	18.496	18.765	16.990	13.337	14.602	15.303
13	14.060	14.788	16.890	19.982	18.248	18.236	16.753	13.187	14.307	15.062
14	13.946	14.635	16.655	19.651	18.000	17.852	16.530	13.046	14.026	14.836
15	13.836	14.489	16.434	19.345	17.757	17.534	16.320	12.912	13.758	14.622
16	13.730	14.348	16.228	19.060	17.522	17.255	16.121	12.786	13.501	14.419
17	13.627	14.213	16.034	18.793	17.297	17.004	15.932	12.667	13.254	14.227
18	13.528	14.083	15.850	18.543	17.083	16.773	15.753	12.553	13.017	14.044
19	13.432	13.957	15.676	18.308	16.882	16.558	15.581	12.444	12.789	13.871
20	13.338	13.835	15.511	18.086	16.692	16.356	15.418	12.341	11.764	13.705
21	13.248	13.717	14.030	17.876	16.513	16.164	15.261	12.241	11.634	13.546
22	13.161	13.602	13.927	17.677	16.345	15.981	15.112	12.146	11.519	13.394
23	13.075	13.490	13.827	17.488	16.187	15.806	14.968	12.055	11.414	13.249
24	12.993	13.381	13.730	17.308	16.037	15.638	14.829	11.967	11.317	13.109
25	12.912	13.274	13.638	17.136	15.895	15.476	14.696	11.882	11.226	12.974
26	12.834	13.170	13.548	16.971	15.760	15.319	14.568	11.801	11.141	12.845
27	12.758	12.943	13.461	16.814	15.631	15.167	14.444	11.722	11.060	12.721
28	12.344	12.866	13.377	16.663	15.509	15.019	14.324	11.646	10.984	12.601
29	12.281	12.792	13.296	16.518	15.392	14.875	14.209	11.572	10.910	12.485
30	12.219	12.720	13.217	16.379	15.280	14.735	14.097	11.501	10.839	12.373
31	12.159	12.650	13.140	16.245	15.172	14.598	13.988	11.431	10.771	12.264
32	12.099	12.582	13.066	16.115	15.068	14.464	13.883	11.364	10.706	12.159
33	12.040	12.516	12.993	15.991	14.969	14.333	13.781	11.299	10.642	12.058
34	11.982	12.452	12.923	15.870	14.873	14.204	13.682	11.236	10.581	11.959
35	11.925	12.389	12.854	15.754	14.780	14.078	13.586	11.174	10.521	11.864
36	11.869	12.328	12.787	15.641	14.690	13.955	13.492	11.114	10.463	11.771
37	11.813	12.268	12.722	15.532	14.603	13.833	13.401	11.056	10.406	11.681
38	11.758	12.210	12.659	15.426	14.518	13.714	13.312	10.999	10.351	11.593
39	11.704	12.153	12.597	15.324	14.436	13.596	13.226	10.944	10.296	11.508
40	11.651	12.097	12.536	15.224	14.357	13.480	13.142	10.890	10.244	11.425
41	11.598	12.043	12.477	15.128	14.279	13.366	13.059	10.837	10.192	11.344
42	11.546	11.989	12.419	15.034	14.204	13.254	12.979	10.786	10.141	11.266
43	11.495	11.937	12.363	14.942	14.130	13.143	12.901	10.735	10.091	11.189
44	11.444	11.886	12.307	14.854	14.058	13.033	12.825	10.686	10.043	11.114
45	11.394	11.836	12.253	14.767	13.988	12.925	12.750	10.638	9.995	11.041
46	11.345	11.787	12.200	14.683	13.920	12.763	12.677	10.591	9.948	10.970
47	11.296	11.739	12.148	14.600	13.853	12.660	12.605	10.545	9.902	10.900
48	11.247	11.692	12.097	14.520	13.788	12.568	12.535	10.500	9.856	10.832
49	11.200	11.646	12.047	14.442	13.724	12.484	12.467	10.456	9.811	10.765

TABLE 2 (continued)

Element	Sb	Te	I	Xe	Cs	Ba	La	Ce	Pr	Nd
P	V	V	V	V	V	V	V	V	V	V
50	11.153	11.601	11.998	14.366	13.661	12.406	12.400	10.413	9.767	10.700
51	11.106	11.557	11.950	14.291	13.599	12.334	12.334	10.370	9.701	10.637
52	11.060	11.513	11.903	14.218	13.539	12.267	12.270	10.329	9.657	10.575
53	11.014	11.470	11.857	14.147	13.480	12.204	12.207	10.288	9.613	10.514
54	10.969	11.428	11.812	14.077	13.422	12.144	12.145	10.248	9.571	10.454
55	10.924	11.387	11.767	14.009	13.365	12.087	12.084	10.209	9.529	10.396
56	10.880	11.346	11.723	13.942	13.309	12.033	12.025	10.170	9.489	10.338
57	10.836	11.306	11.680	13.877	13.254	11.981	11.966	10.132	9.450	10.282
58	10.793	11.267	11.637	13.813	13.200	11.931	11.909	10.095	9.413	10.227
59	10.750	11.228	11.596	13.750	13.147	11.883	11.852	10.058	9.376	10.173
60	10.708	11.190	11.555	13.689	13.094	11.837	11.797	10.022	9.341	10.120
61	10.666	11.153	11.515	13.409	13.043	11.793	11.743	9.987	9.306	10.068
62	10.624	11.116	11.476	13.348	12.992	11.750	11.689	9.952	9.273	10.017
63	10.583	11.079	11.437	13.288	12.942	11.708	11.637	9.918	9.241	9.967
64	10.542	11.044	11.398	13.229	12.893	11.667	11.585	9.884	9.210	9.918
65	10.502	11.009	11.361	13.171	12.844	11.628	11.534	9.851	9.180	9.869
66	10.462	10.974	11.323	13.115	12.797	11.589	11.484	9.819	9.151	9.822
67	10.422	10.940	11.287	13.059	12.749	11.552	11.435	9.787	9.122	9.775
68	10.383	10.906	11.251	13.004	12.703	11.516	11.387	9.755	9.095	9.729
69	10.344	10.873	11.215	12.951	12.657	11.480	11.339	9.724	9.068	9.684
70	10.305	10.840	11.180	12.898	12.612	11.446	11.292	9.693	9.042	9.640
71	10.267	10.808	11.146	12.846	12.092	11.412	11.246	9.663	9.017	9.596
72	10.229	10.776	11.111	12.795	12.054	11.378	11.200	9.633	8.992	9.553
73	10.191	10.745	11.078	12.745	12.018	11.346	11.155	9.604	8.968	9.511
74	10.154	10.714	11.044	12.696	11.982	11.314	11.111	9.575	8.945	9.469
75	10.117	10.683	11.012	12.648	11.946	11.283	11.068	9.546	8.922	9.428
76	10.080	10.653	10.979	12.600	11.911	11.252	11.025	9.518	8.900	9.388
77	10.044	10.623	10.948	12.553	11.877	11.222	10.982	9.490	8.878	9.348
78	10.008	10.594	10.916	12.507	11.843	11.193	10.941	9.463	8.857	9.309
79	9.972	10.565	10.886	12.462	11.810	11.164	10.899	9.436	8.836	9.271
80	9.936	10.536	10.855	12.417	11.777	11.136	10.859	9.409	8.816	9.233
81	9.901	10.508	10.825	12.373	11.744	11.108	10.819	9.383	8.796	9.130
82	9.866	10.480	10.795	12.329	11.712	11.080	10.779	9.357	8.776	9.093
83	9.831	10.452	10.765	12.287	11.681	11.053	10.740	9.331	8.757	9.058
84	9.796	10.425	10.736	12.245	11.650	11.027	10.702	9.306	8.739	9.022
85	9.762	10.398	10.707	12.203	11.619	11.000	10.664	9.281	8.720	8.987
86	9.728	10.371	10.679	12.162	11.588	10.974	10.626	9.256	8.702	8.953
87	9.694	10.345	10.651	12.122	11.558	10.949	10.589	9.232	8.685	8.919
88	9.661	10.319	10.623	12.082	11.529	10.924	10.553	9.208	8.668	8.886
89	9.627	10.293	10.596	12.043	11.499	10.899	10.517	9.184	8.651	8.852
90	9.594	10.268	10.569	12.004	11.470	10.875	10.481	9.160	8.634	8.820
91	9.561	10.243	10.542	11.966	11.442	10.851	10.446	9.137	8.617	8.788
92	9.528	10.218	10.515	11.928	11.413	10.827	10.411	9.114	8.601	8.756
93	9.496	10.193	10.489	11.891	11.385	10.803	10.376	9.091	8.585	8.725
94	9.463	10.169	10.463	11.855	11.357	10.780	10.342	9.069	8.570	8.694
95	9.431	10.145	10.438	11.818	11.330	10.757	10.309	9.047	8.554	8.663
96	9.399	10.121	10.412	11.783	11.303	10.735	10.276	9.025	8.539	8.633
97	9.368	10.097	10.387	11.747	11.276	10.713	10.243	9.003	8.524	8.603
98	9.336	10.074	10.363	11.713	11.249	10.690	10.210	8.981	8.510	8.574
99	9.305	10.051	10.338	11.678	11.222	10.669	10.178	8.960	8.495	8.545
100	9.273	10.028	10.314	11.644	11.196	10.647	10.147	8.939	8.481	8.516

TABLE 2 (continued)

Element	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
P	V	V	V	V	V	V	V	V	V	V
0	20.133	19.902	28.383	19.795	19.118	18.873	18.599	18.309	17.964	24.586
1	19.597	19.508	26.266	19.017	18.601	18.389	18.145	17.884	17.468	22.849
2	19.091	19.097	24.633	18.361	18.137	17.954	17.734	17.498	17.028	21.571
3	18.610	18.673	23.304	17.796	17.715	17.557	17.357	17.145	16.633	20.489
4	18.153	18.247	22.184	17.299	17.329	17.194	17.011	16.819	16.275	19.567
5	17.720	17.827	21.215	16.858	16.972	16.859	16.690	16.517	15.947	18.774
6	17.314	17.422	20.362	16.460	16.642	16.548	16.391	16.235	15.646	18.085
7	16.934	17.037	19.599	16.099	16.334	16.258	16.111	15.972	15.368	17.481
8	16.579	16.675	18.908	15.768	16.046	15.987	15.849	15.725	15.108	16.945
9	16.250	16.337	18.274	15.464	15.775	15.732	15.602	15.491	14.866	16.467
10	15.943	16.021	17.689	15.182	15.519	15.491	15.369	15.271	14.639	16.037
11	15.657	15.727	17.144	14.920	15.277	15.264	15.147	15.062	14.426	15.648
12	15.390	15.452	16.631	14.675	15.048	15.048	14.937	14.863	14.224	15.293
13	15.139	15.194	15.950	14.445	14.830	14.843	14.738	14.674	14.033	14.968
14	14.903	14.951	15.610	14.229	14.622	14.647	14.547	14.493	13.852	14.668
15	14.681	14.723	15.220	14.025	14.423	14.461	14.365	14.321	13.679	14.391
16	14.471	14.507	14.804	13.832	14.233	14.282	14.191	14.155	13.515	14.134
17	14.271	14.302	14.408	13.648	14.051	14.111	14.024	13.996	13.358	13.894
18	14.081	14.107	14.060	13.474	13.876	13.947	13.863	13.844	13.208	13.670
19	13.900	13.921	13.762	13.308	13.708	13.789	13.709	13.697	13.064	13.459
20	13.727	13.743	13.506	13.150	13.546	13.637	13.561	13.556	12.926	13.262
21	13.562	13.573	13.284	12.999	13.390	13.491	13.418	13.419	12.794	13.075
22	13.403	13.410	13.087	12.853	13.239	13.350	13.280	13.288	12.666	12.899
23	13.251	13.253	12.911	12.714	13.093	13.213	13.146	13.160	12.543	12.732
24	13.104	13.101	12.751	12.581	12.952	13.082	13.017	13.037	12.425	12.573
25	12.963	12.955	12.605	12.452	12.816	12.954	12.893	12.918	12.310	12.422
26	12.827	12.814	12.469	12.328	12.683	12.830	12.772	12.802	12.200	12.278
27	12.695	12.677	12.343	12.209	12.554	12.710	12.655	12.690	12.093	12.141
28	12.568	12.545	12.225	12.094	12.429	12.594	12.541	12.581	11.989	12.010
29	12.445	12.416	12.114	11.983	12.308	12.481	12.431	12.475	11.889	11.884
30	12.326	12.292	12.009	11.875	12.190	12.371	12.324	12.372	11.791	11.764
31	12.209	12.170	11.909	11.771	12.075	12.264	12.219	12.272	11.697	11.648
32	12.096	12.052	11.814	11.670	11.962	12.160	12.118	12.175	11.605	11.537
33	11.987	11.937	11.723	11.573	11.853	12.059	12.019	12.080	11.516	11.431
34	11.880	11.825	11.636	11.478	11.746	11.960	11.923	11.987	11.429	11.328
35	11.777	11.715	11.552	11.386	11.642	11.864	11.830	11.897	11.344	11.229
36	11.675	11.608	11.472	11.296	11.540	11.770	11.738	11.808	11.262	11.133
37	11.577	11.504	11.395	11.209	11.440	11.678	11.649	11.722	11.182	11.041
38	11.481	11.402	11.320	11.125	11.343	11.588	11.562	11.638	11.104	10.951
39	11.387	11.302	11.247	11.042	11.248	11.501	11.478	11.556	11.028	10.865
40	11.295	11.204	11.177	10.962	11.154	11.415	11.395	11.475	10.953	10.781
41	11.204	11.105	11.109	10.884	11.063	11.331	11.314	11.397	10.881	10.700
42	11.114	11.006	11.043	10.808	10.973	11.249	11.235	11.320	10.810	10.621
43	11.028	10.915	10.979	10.734	10.885	11.169	11.157	11.244	10.740	10.545
44	10.948	10.831	10.916	10.661	10.799	11.090	11.081	11.171	10.672	10.471
45	10.872	10.753	10.855	10.590	10.714	11.013	11.007	11.098	10.606	10.399
46	10.799	10.679	10.795	10.521	10.630	10.937	10.934	11.027	10.541	10.329
47	10.729	10.609	10.737	10.453	10.549	10.863	10.863	10.958	10.478	10.261
48	10.662	10.542	10.681	10.387	10.468	10.791	10.793	10.890	10.416	10.195
49	10.597	10.478	10.625	10.323	10.389	10.719	10.725	10.823	10.355	10.130

TABLE 2 (continued)

Element	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
P	V	V	V	V	V	V	V	V	V	V
50	10.534	10.417	10.571	10.259	10.311	10.649	10.658	10.757	10.295	10.067
51	10.474	10.358	10.518	10.197	9.842	10.580	10.592	10.692	10.236	10.006
52	10.414	10.301	10.466	10.137	9.810	10.513	10.528	10.629	10.179	9.946
53	10.357	10.246	10.415	10.077	9.780	10.446	10.465	10.567	10.123	9.888
54	10.301	10.193	10.365	10.019	9.749	10.381	10.402	10.506	10.067	9.831
55	10.247	10.141	10.316	9.962	9.719	10.317	10.341	10.446	10.013	9.775
56	10.193	10.090	10.267	9.906	9.690	10.254	10.282	10.386	9.960	9.721
57	10.141	10.041	10.220	9.851	9.660	10.192	10.223	10.328	9.908	9.668
58	10.090	9.994	10.174	9.798	9.631	10.130	10.165	10.271	9.856	9.616
59	10.041	9.947	10.128	9.745	9.602	10.070	10.108	10.215	9.806	9.565
60	9.992	9.902	10.083	9.693	9.574	10.011	10.052	10.159	9.756	9.515
61	9.944	9.857	10.039	9.644	9.546	9.953	9.997	10.105	9.708	9.466
62	9.898	9.814	9.995	9.631	9.518	9.895	9.943	10.051	9.660	9.419
63	9.852	9.771	9.952	9.298	9.490	9.839	9.890	9.998	9.613	9.372
64	9.807	9.729	9.910	9.266	9.463	9.783	9.838	9.946	9.566	9.326
65	9.763	9.688	9.868	9.235	9.436	9.728	9.786	9.895	9.521	9.282
66	9.720	9.648	9.827	9.204	9.409	9.673	9.736	9.844	9.476	9.238
67	9.678	9.609	9.787	9.173	9.382	9.620	9.686	9.794	9.432	9.195
68	9.636	9.570	9.747	9.143	9.356	9.567	9.637	9.745	9.388	9.152
69	9.595	9.532	9.708	9.113	9.330	9.515	9.588	9.696	9.345	9.111
70	9.554	9.494	9.669	9.084	9.304	9.463	9.541	9.649	9.303	9.070
71	9.515	9.457	9.631	9.055	9.278	9.413	9.494	9.601	9.262	9.030
72	9.476	9.421	9.593	9.026	9.253	9.363	9.447	9.555	9.221	8.991
73	9.437	9.385	9.556	8.998	9.228	9.313	9.402	9.509	9.180	8.952
74	9.399	9.350	9.519	8.970	9.202	8.645	9.357	9.463	9.141	8.914
75	9.362	9.315	9.482	8.943	9.178	8.608	9.313	9.419	9.102	8.877
76	9.325	9.281	9.446	8.915	9.153	8.573	9.269	9.374	9.063	8.841
77	9.289	9.247	9.411	8.889	9.128	8.539	9.226	9.331	9.025	8.805
78	9.253	9.214	9.376	8.862	9.104	8.507	9.183	9.288	8.987	8.769
79	9.218	9.181	9.341	8.836	9.080	8.475	9.142	9.245	8.950	8.734
80	9.184	9.149	9.306	8.810	9.056	8.445	9.100	9.203	8.914	8.700
81	9.122	9.117	9.273	8.784	9.032	8.415	9.059	9.161	8.878	8.667
82	9.088	9.085	9.239	8.759	9.009	8.386	9.019	9.120	8.842	8.633
83	9.056	9.054	9.206	8.734	8.985	8.358	8.979	9.080	8.807	8.601
84	9.023	9.023	9.173	8.709	8.962	8.331	8.940	9.040	8.772	8.569
85	8.990	8.992	9.140	8.685	8.939	8.304	8.901	9.000	8.738	8.537
86	8.958	8.962	9.108	8.661	8.916	8.278	8.863	8.961	8.704	8.506
87	8.927	8.932	9.076	8.637	8.893	8.252	8.826	8.922	8.671	8.475
88	8.895	8.902	9.044	8.613	8.871	8.227	8.788	8.883	8.638	8.445
89	8.864	8.873	9.013	8.590	8.848	8.202	8.751	8.845	8.605	8.415
90	8.834	8.844	8.982	8.567	8.826	8.178	8.715	8.808	8.573	8.386
91	8.804	8.816	8.951	8.544	8.804	8.155	8.679	8.771	8.541	8.357
92	8.774	8.787	8.921	8.521	8.781	8.131	8.644	8.734	8.510	8.328
93	8.745	8.759	8.891	8.499	8.760	8.108	8.609	8.697	8.479	8.300
94	8.716	8.731	8.861	8.477	8.738	8.086	8.574	8.661	8.448	8.272
95	8.687	8.704	8.832	8.455	8.716	8.064	8.540	8.626	8.418	8.245
96	8.658	8.676	8.802	8.433	8.694	8.042	8.506	8.590	8.388	8.218
97	8.630	8.649	8.773	8.412	8.673	8.021	8.472	8.555	8.358	8.191
98	8.603	8.623	8.744	8.391	8.652	7.999	8.439	8.521	8.329	8.165
99	8.575	8.596	8.716	8.370	8.631	7.979	8.406	8.486	8.300	8.139
100	8.548	8.570	8.687	8.349	8.610	7.958	8.374	8.452	8.271	8.114

TABLE 2 (continued)

Element	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
P	V	V	V	V	V	V	V	V	V	V
0	17.650	13.381	10.804	9.524	8.830	8.397	8.487	9.034	10.110	13.580
1	17.326	13.269	10.750	9.493	8.806	8.377	8.465	9.003	10.054	13.274
2	17.019	13.162	10.697	9.462	8.782	8.357	8.444	8.973	10.000	13.063
3	16.729	13.057	10.645	9.431	8.758	8.338	8.423	8.943	9.948	12.877
4	16.453	12.956	10.594	9.401	8.735	8.318	8.402	8.913	9.898	12.709
5	16.190	12.858	10.545	9.372	8.712	8.299	8.382	8.885	9.850	12.555
6	15.941	12.763	10.496	9.343	8.689	8.280	8.361	8.856	9.803	12.411
7	15.703	12.670	10.448	9.315	8.667	8.262	8.341	8.829	9.757	12.276
8	15.476	12.580	10.402	9.287	8.645	8.243	8.321	8.802	9.713	12.148
9	15.259	12.492	10.356	9.259	8.623	8.225	8.301	8.775	9.669	12.027
10	15.052	12.407	10.311	9.232	8.602	8.207	8.282	8.749	9.628	11.738
11	14.853	12.324	10.266	9.205	8.580	8.189	8.263	8.723	9.587	11.637
12	14.663	12.242	10.223	9.178	8.560	8.171	8.244	8.697	9.547	11.544
13	14.480	12.163	10.180	9.152	8.539	8.154	8.225	8.672	9.508	11.457
14	14.305	12.086	10.139	9.127	8.519	8.136	8.206	8.648	9.471	11.375
15	14.136	12.010	10.097	9.101	8.498	8.119	8.187	8.624	9.434	11.297
16	13.974	11.936	10.057	9.077	8.479	8.102	8.169	8.600	9.398	11.223
17	13.818	11.864	10.017	9.052	8.459	8.086	8.151	8.576	9.363	11.153
18	13.667	11.793	9.978	9.028	8.440	8.069	8.133	8.553	9.329	11.085
19	13.522	11.723	9.940	9.004	8.421	8.052	8.115	8.531	9.295	11.019
20	13.382	11.656	9.902	8.980	8.402	8.036	8.098	8.508	9.262	10.956
21	13.247	11.589	9.865	8.957	8.383	8.020	8.080	8.486	9.230	10.895
22	13.115	11.524	9.828	8.934	8.364	8.004	8.063	8.464	9.198	10.836
23	12.989	11.460	9.792	8.911	8.346	7.988	8.046	8.443	9.168	10.778
24	12.866	11.397	9.756	8.888	8.328	7.972	8.029	8.422	9.137	10.722
25	12.747	11.336	9.721	8.866	8.310	7.957	8.012	8.401	9.108	10.667
26	12.632	11.276	9.686	8.844	8.293	7.941	7.996	8.380	9.078	10.614
27	12.520	11.216	9.652	8.822	8.275	7.926	7.979	8.360	9.050	10.562
28	12.411	11.158	9.619	8.801	8.258	7.911	7.963	8.340	9.022	10.511
29	12.306	11.101	9.586	8.780	8.241	7.896	7.947	8.320	8.994	10.461
30	12.203	11.045	9.553	8.759	8.224	7.881	7.931	8.301	8.967	10.412
31	12.103	10.990	9.521	8.738	8.207	7.867	7.915	8.281	8.941	10.364
32	12.006	10.935	9.489	8.718	8.190	7.852	7.899	8.262	8.915	10.317
33	11.912	10.882	9.458	8.698	8.174	7.837	7.884	8.243	8.889	10.271
34	11.820	10.829	9.427	8.678	8.158	7.823	7.868	8.225	8.864	10.226
35	11.730	10.777	9.397	8.658	8.142	7.809	7.853	8.206	8.839	10.181
36	11.643	10.727	9.366	8.638	8.126	7.795	7.838	8.188	8.814	10.098
37	11.558	10.676	9.337	8.619	8.110	7.781	7.822	8.170	8.790	10.053
38	11.474	10.627	9.307	8.600	8.094	7.767	7.808	8.153	8.766	10.009
39	11.393	10.578	9.278	8.581	8.079	7.753	7.793	8.135	8.743	9.967
40	11.314	10.530	9.250	8.562	8.064	7.740	7.778	8.118	8.720	9.927
41	11.237	10.483	9.221	8.544	8.048	7.726	7.763	8.101	8.697	9.889
42	11.161	10.437	9.193	8.525	8.033	7.713	7.749	8.084	8.675	9.852
43	11.087	10.391	9.166	8.507	8.018	7.699	7.735	8.067	8.653	9.815
44	11.015	10.346	9.139	8.489	8.004	7.686	7.720	8.051	8.631	9.780
45	10.944	10.301	9.112	8.471	7.989	7.673	7.706	8.034	8.610	9.746
46	10.875	10.257	9.085	8.453	7.975	7.660	7.692	8.018	8.589	9.713
47	10.807	10.213	9.059	8.436	7.960	7.647	7.679	8.002	8.568	9.681
48	10.741	10.171	9.033	8.419	7.946	7.635	7.665	7.986	8.547	9.649
49	10.676	10.128	9.007	8.402	7.932	7.622	7.651	7.970	8.527	9.618

TABLE 2 (continued)

Element	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
P	V	V	V	V	V	V	V	V	V	V
50	10.612	10.087	8.981	8.385	7.918	7.609	7.638	7.955	8.507	9.588
51	10.550	9.885	8.956	8.368	7.904	7.597	7.624	7.939	8.487	9.559
52	10.489	9.830	8.931	8.351	7.890	7.584	7.611	7.924	8.468	9.530
53	10.429	9.775	8.907	8.335	7.877	7.572	7.597	7.909	8.449	9.501
54	10.370	9.720	8.882	8.318	7.863	7.560	7.584	7.894	8.430	9.473
55	10.313	9.665	8.858	8.302	7.850	7.548	7.571	7.879	8.411	9.446
56	10.256	9.610	8.834	8.286	7.836	7.536	7.558	7.865	8.392	9.419
57	10.200	9.555	8.810	8.270	7.823	7.524	7.545	7.850	8.374	9.392
58	10.146	9.500	8.787	8.254	7.810	7.512	7.533	7.836	8.356	9.366
59	10.092	9.445	8.764	8.238	7.797	7.500	7.520	7.822	8.338	9.341
60	10.040	9.390	8.741	8.223	7.784	7.489	7.507	7.808	8.320	9.316
61	9.988	9.335	8.718	8.208	7.772	7.477	7.495	7.794	8.303	9.291
62	9.937	9.280	8.696	8.192	7.759	7.465	7.482	7.780	8.286	9.266
63	9.887	9.225	8.674	8.177	7.746	7.454	7.470	7.766	8.269	9.242
64	9.838	9.170	8.652	8.162	7.734	7.443	7.458	7.752	8.252	9.218
65	9.790	9.149	8.630	8.147	7.722	7.431	7.446	7.739	8.235	9.195
66	9.743	9.125	8.608	8.133	7.709	7.420	7.434	7.726	8.218	9.171
67	9.696	9.100	8.587	8.118	7.697	7.409	7.422	7.712	8.202	9.148
68	9.650	9.076	8.565	8.103	7.685	7.398	7.410	7.699	8.186	9.126
69	9.605	9.051	8.544	8.089	7.673	7.387	7.398	7.686	8.170	9.103
70	9.560	9.027	8.524	8.075	7.661	7.376	7.386	7.673	8.154	9.081
71	9.516	9.002	8.503	8.061	7.649	7.365	7.375	7.660	8.138	9.059
72	9.473	8.978	8.483	8.047	7.638	7.355	7.363	7.648	8.123	9.038
73	9.431	8.953	8.462	8.033	7.626	7.344	7.352	7.635	8.107	9.016
74	9.389	8.929	8.442	8.019	7.614	7.333	7.340	7.623	8.092	8.995
75	9.348	8.904	8.422	8.005	7.603	7.323	7.329	7.610	8.077	8.974
76	9.307	8.880	8.403	7.991	7.591	7.312	7.318	7.598	8.062	8.953
77	9.267	8.855	8.383	7.978	7.580	7.302	7.306	7.586	8.047	8.933
78	9.228	8.831	8.364	7.965	7.569	7.292	7.295	7.574	8.033	8.912
79	9.189	8.806	8.344	7.951	7.558	7.281	7.284	7.562	8.018	8.892
80	9.151	8.782	8.325	7.938	7.547	7.271	7.273	7.550	8.004	8.872
81	9.113	8.757	8.307	7.925	7.536	7.261	7.262	7.538	7.990	8.853
82	9.076	8.733	8.288	7.912	7.525	7.251	7.251	7.526	7.976	8.833
83	9.039	8.708	8.269	7.899	7.514	7.241	7.241	7.515	7.962	8.814
84	9.003	8.684	8.251	7.886	7.503	7.231	7.230	7.503	7.948	8.794
85	8.967	8.659	8.233	7.873	7.492	7.221	7.219	7.492	7.934	8.775
86	8.932	8.635	8.214	7.861	7.482	7.211	7.209	7.481	7.921	8.756
87	8.897	8.610	8.196	7.848	7.471	7.202	7.198	7.469	7.907	8.738
88	8.863	8.586	8.179	7.836	7.461	7.192	7.188	7.458	7.894	8.719
89	8.829	8.561	8.161	7.823	7.450	7.182	7.177	7.447	7.881	8.701
90	8.795	8.537	8.143	7.811	7.440	7.173	7.167	7.436	7.868	8.682
91	8.762	8.512	8.126	7.799	7.430	7.163	7.157	7.425	7.855	8.664
92	8.730	8.488	8.109	7.787	7.419	7.154	7.146	7.414	7.842	8.646
93	8.204	8.463	8.092	7.775	7.409	7.144	7.136	7.403	7.829	8.628
94	8.190	8.439	8.075	7.763	7.399	7.135	7.126	7.393	7.816	8.610
95	8.176	8.414	8.058	7.751	7.389	7.126	7.116	7.382	7.804	8.593
96	8.163	8.390	8.041	7.739	7.379	7.116	7.106	7.371	7.791	8.575
97	8.149	8.365	8.024	7.727	7.369	7.107	7.096	7.361	7.779	8.558
98	8.135	8.341	8.008	7.716	7.359	7.098	7.087	7.351	7.767	8.541
99	8.121	8.316	7.992	7.704	7.350	7.089	7.077	7.340	7.755	8.524
100	8.108	8.291	7.975	7.692	7.340	7.080	7.067	7.330	7.743	8.507

TABLE 2 (continued)

Element	Tl	Pb	Bi	Po	At	Rn	Fr	Ra	Ac	Th
P	V	V	V	V	V	V	V	V	V	V
0	16.862	17.881	21.084	22.432	22.935	49.753	64.449	40.837	26.523	19.640
1	16.452	17.529	20.538	21.981	22.165	36.613	47.902	36.339	25.557	19.307
2	16.098	17.213	20.075	21.578	21.526	32.450	41.941	33.626	24.747	18.997
3	15.787	16.926	17.939	21.215	20.983	30.039	37.536	31.614	24.044	18.709
4	15.509	16.664	17.667	20.883	20.513	28.372	34.118	30.010	23.420	18.439
5	15.258	16.423	17.423	20.579	20.099	27.109	31.470	28.683	22.857	18.185
6	15.030	16.199	17.201	20.297	19.729	26.100	29.420	27.558	22.344	17.945
7	14.820	15.991	16.995	20.035	19.395	25.262	27.814	26.587	21.872	17.719
8	14.626	15.796	16.803	19.791	19.092	24.547	26.530	25.738	21.437	17.504
9	14.446	15.613	16.185	19.561	18.813	23.926	25.479	24.986	21.032	17.300
10	14.278	15.440	16.004	19.345	18.555	23.378	24.599	24.316	20.653	17.106
11	14.121	15.278	15.834	19.140	18.316	22.888	23.850	23.712	20.299	16.920
12	13.972	15.123	15.673	18.947	18.092	22.445	23.200	23.165	19.967	16.742
13	13.833	14.977	15.522	18.762	17.883	22.042	22.630	22.666	19.653	16.572
14	13.700	14.834	15.379	18.587	17.686	21.672	22.123	22.209	19.357	16.409
15	13.574	14.710	15.243	18.419	17.500	21.330	21.668	21.788	19.077	16.252
16	13.455	14.591	15.114	18.259	17.323	21.013	21.256	21.398	18.812	16.101
17	13.341	14.477	14.990	18.105	17.156	20.718	20.880	21.036	18.560	15.955
18	13.232	14.368	14.872	17.957	16.996	20.442	20.535	20.698	18.320	15.815
19	13.127	14.263	14.759	17.815	16.844	20.182	20.216	20.382	18.092	15.679
20	13.027	14.162	14.650	17.679	16.698	19.937	19.921	20.086	17.874	15.548
21	12.931	14.064	14.546	17.547	16.559	19.706	19.646	19.807	17.665	15.421
22	12.839	13.970	14.446	17.420	16.425	19.486	19.388	19.544	17.466	15.299
23	12.750	13.879	14.349	17.297	16.297	19.278	19.146	19.295	17.276	15.179
24	12.664	13.791	14.255	17.178	16.173	19.080	18.918	19.059	17.093	15.064
25	12.581	13.706	14.165	17.062	16.053	18.890	18.703	18.834	16.918	14.952
26	12.501	13.624	14.078	16.950	15.938	18.709	18.499	18.621	16.749	14.843
27	12.424	13.544	13.994	16.842	15.827	18.536	18.306	18.418	16.587	14.737
28	12.349	13.466	13.912	16.737	15.719	18.370	18.122	18.224	16.431	14.634
29	12.276	13.391	13.832	16.634	15.615	18.210	17.946	18.038	16.281	14.533
30	12.206	13.318	13.755	16.535	15.514	18.057	17.778	17.860	16.136	14.436
31	12.138	13.247	13.681	16.438	15.416	17.909	17.618	17.689	15.997	14.340
32	12.071	13.177	13.608	16.343	15.321	17.766	17.464	17.526	15.862	14.247
33	12.007	13.110	13.537	16.251	15.228	17.629	17.317	17.368	15.731	14.157
34	11.944	13.044	13.468	16.161	15.138	17.496	17.175	17.216	15.605	14.068
35	11.883	12.980	13.401	16.074	15.051	17.367	17.038	17.070	15.483	13.982
36	11.823	12.917	13.336	15.988	14.966	17.243	16.907	16.929	15.365	13.897
37	11.765	12.856	13.272	15.904	14.883	17.122	16.780	16.793	15.251	13.814
38	11.709	12.796	13.210	15.823	14.802	17.005	16.657	16.662	15.140	13.734
39	11.653	12.738	13.149	15.743	14.723	16.892	16.539	16.534	15.032	13.655
40	11.600	12.681	13.089	15.665	14.645	16.781	16.424	16.411	14.927	13.577
41	11.547	12.625	13.031	15.588	14.570	16.674	16.313	16.292	14.826	13.501
42	11.495	12.571	12.974	15.513	14.496	16.570	16.205	16.176	14.727	13.427
43	11.445	12.517	12.919	15.440	14.424	16.468	16.101	16.063	14.631	13.354
44	11.396	12.465	12.864	15.368	14.354	16.370	15.999	15.954	14.538	13.283
45	11.348	12.414	12.811	15.297	14.285	16.273	15.901	15.848	14.447	13.213
46	11.301	12.364	12.759	15.228	14.218	16.179	15.805	15.745	14.358	13.145
47	11.255	12.314	12.708	15.160	14.151	16.088	15.712	15.645	14.272	13.077
48	11.209	12.266	12.658	15.094	14.087	15.998	15.621	15.547	14.188	13.011
49	11.165	12.219	12.609	15.029	14.023	15.911	15.533	15.452	14.106	12.946

TABLE 2 (continued)

Element	Tl	Pb	Bi	Po	At	Rn	Fr	Ra	Ac	Th
P	V	V	V	V	V	V	V	V	V	V
50	11.122	12.173	12.560	14.964	13.961	15.826	15.446	15.359	14.026	12.883
51	11.079	12.127	12.513	14.901	13.900	15.743	15.362	15.269	13.947	12.820
52	11.037	12.082	12.467	14.839	13.840	15.661	15.280	15.181	13.871	12.759
53	10.996	12.038	12.421	14.779	13.781	15.581	15.200	15.095	13.797	12.698
54	10.956	11.995	12.376	14.719	13.723	15.503	15.122	15.011	13.724	12.639
55	10.917	11.953	12.332	14.660	13.667	15.427	15.046	14.929	13.652	12.581
56	10.878	11.911	12.289	14.602	13.611	15.352	14.971	14.848	13.583	12.523
57	10.840	11.870	12.247	14.545	13.556	15.279	14.898	14.770	13.515	12.467
58	10.802	11.830	12.205	14.489	13.502	15.207	14.826	14.693	13.448	12.411
59	10.766	11.790	12.164	14.434	13.449	15.136	14.756	14.618	13.383	12.357
60	10.729	11.751	12.123	14.380	13.397	15.067	14.688	14.544	13.319	12.303
61	10.694	11.713	12.083	14.326	13.346	15.000	14.621	14.472	13.256	12.250
62	10.659	11.675	12.044	14.274	13.296	14.933	14.555	14.402	13.195	12.197
63	10.624	11.638	12.006	14.222	13.246	14.868	14.491	14.333	13.135	12.146
64	10.590	11.602	11.968	14.170	13.197	14.804	14.427	14.265	13.076	12.095
65	10.557	11.566	11.930	14.120	13.149	14.741	14.365	14.199	13.018	12.046
66	10.524	11.530	11.894	14.070	13.102	14.679	14.304	14.134	12.961	11.996
67	10.492	11.495	11.857	14.021	13.055	14.618	14.245	14.070	12.905	11.948
68	10.460	11.461	11.821	13.973	13.009	14.558	14.186	14.007	12.851	11.900
69	10.429	11.427	11.786	13.926	12.964	14.499	14.128	13.945	12.797	11.853
70	10.398	11.393	11.751	13.879	12.919	14.442	14.072	13.885	12.744	11.806
71	10.367	11.360	11.717	13.832	12.875	14.385	14.016	13.826	12.693	11.761
72	10.337	11.328	11.683	13.786	12.832	14.329	13.961	13.767	12.642	11.715
73	10.308	11.296	11.650	13.741	12.789	14.274	13.908	13.710	12.592	11.671
74	10.278	11.264	11.617	13.697	12.747	14.220	13.855	13.654	12.543	11.627
75	10.250	11.233	11.585	13.653	12.705	14.166	13.803	13.598	12.494	11.583
76	10.221	11.202	11.553	13.609	12.664	14.114	13.752	13.544	12.447	11.540
77	10.193	11.171	11.521	13.566	12.623	14.062	13.701	13.490	12.400	11.498
78	10.166	11.141	11.490	13.524	12.583	14.011	13.652	13.438	12.354	11.456
79	10.138	11.112	11.459	13.482	12.543	13.961	13.603	13.386	12.309	11.415
80	10.111	11.082	11.429	13.441	12.504	13.911	13.555	13.335	12.264	11.374
81	10.085	11.054	11.399	13.400	12.466	13.862	13.508	13.284	12.220	11.334
82	10.059	11.025	11.369	13.360	12.428	13.814	13.461	13.235	12.177	11.294
83	10.033	10.997	11.340	13.320	12.390	13.767	13.415	13.186	12.135	11.255
84	10.007	10.969	11.311	13.280	12.353	13.720	13.370	13.138	12.093	11.216
85	9.982	10.941	11.282	13.241	12.316	13.674	13.326	13.091	12.052	11.178
86	9.957	10.914	11.254	13.203	12.280	13.629	13.282	13.044	12.011	11.140
87	9.932	10.887	11.226	13.165	12.244	13.584	13.238	12.998	11.971	11.102
88	9.908	10.861	11.198	13.127	12.209	13.539	13.196	12.953	11.932	11.065
89	9.884	10.835	11.171	13.090	12.174	13.496	13.153	12.908	11.893	11.029
90	9.860	10.809	11.144	13.053	12.139	13.452	13.112	12.865	11.854	10.993
91	9.837	10.783	11.117	13.016	12.105	13.410	13.071	12.821	11.817	10.957
92	9.813	10.758	11.091	12.980	12.071	13.368	13.030	12.778	11.779	10.921
93	9.790	10.733	11.065	12.945	12.037	13.326	12.990	12.736	11.743	10.886
94	9.768	10.708	11.039	12.909	12.004	13.285	12.951	12.694	11.706	10.852
95	9.745	10.683	11.013	12.874	11.971	13.245	12.912	12.653	11.671	10.817
96	9.723	10.659	10.988	12.840	11.939	13.205	12.874	12.613	11.635	10.784
97	9.701	10.635	10.963	12.806	11.907	13.165	12.836	12.573	11.600	10.750
98	9.679	10.611	10.938	12.772	11.875	13.126	12.798	12.533	11.566	10.717
99	9.658	10.588	10.914	12.738	11.844	13.087	12.761	12.494	11.532	10.684
100	9.637	10.565	10.890	12.705	11.812	13.049	12.725	12.456	11.499	10.651

TABLE 3
Comparison of $T = 0$ K atomic volumes (in m^3/Mmol) at 100 GPa from this work and other tabulations²²⁶⁻²²⁹.

Element	This Work (0 K)	McQueen (0 K)	Kennedy (298 K)	Al'tshuler (293 K)	SESAME (≤ 30 K)	Shock Reduction (This Work)
H	1.417				1.53	
He	2.077				2.19	
Li	3.612				3.93	
Be	3.150				3.20	3.20
B	3.321				3.19	
C	2.906				2.91	
N	4.126				4.33	
O	3.484				4.09	
F	3.334				4.31	
Ne	3.929				3.91	
Na	6.272				6.49	
Mg	6.777			6.84	6.83	
Al	6.264			6.24	6.15	6.03
Si	5.458				5.55	
P	5.844					
S	5.931					
Cl	7.381					
Ar	7.613					
K	6.946				6.92	
Ca	8.121				7.41	
Sc	7.327					
Ti	6.787		6.68			
V	5.969	5.94	5.91	5.91	5.92	5.90
Cr	5.534	5.59	5.48		5.50	5.50
Mn	5.352					
Fe	5.048				5.10	
Co	5.022		5.01	5.00	4.99	5.00
Ni	4.977		4.96	5.00	4.97	4.98
Cu	5.073		5.08	5.12	5.11	5.12
Zn	5.711		5.69	5.75	5.79	

TABLE 3 (continued)

Element	This Work (0 K)	McQueen (0 K)	Kennedy (298 K)	Al'tshuler (293 K)	SESAME (≤ 30 K)	Shock Reduction (This Work)
Ga	6.490				6.75	
Ge	6.947				6.76	
As	7.093					
Se	7.403					
Br	8.550					
Kr	9.454				9.98	
Rb	9.155				8.46	
Sr	8.999					
Y	8.687				8.80	
Zr	8.404		8.25		8.31	
Nb	7.699	7.80	7.81	7.83	7.80	7.77
Mo	7.370	7.40	7.38	7.41	7.40	7.40
Tc	7.073					
Ru	6.642					
Rh	6.638	6.65		6.67	6.64	6.63
Pd	6.781	6.79	6.77	6.85	6.77	6.78
Ag	7.154		7.13	7.18	7.21	7.14
Cd	7.870		7.79	7.95	8.12	
In	8.950			9.02	8.92	
Sn	9.424				9.20	
Sb	9.273				9.90	
Te	10.028					
I	10.314					
Xe	11.644				11.78	
Cs	11.196				11.22	
Ba	10.647					
La	10.147					
Ce	8.939					
Pr	8.481					
Nd	8.516					

TABLE 3 (continued)

Element	This Work (0 K)	McQueen (0 K)	Kennedy (298 K)	Al'tshuler (293 K)	SESAME (≤ 30 K)	Shock Reduction (This Work)
Pm	8.548					
Sm	8.570					
Eu	8.687				9.81	
Gd	8.349					
Tb	8.610					
Dy	7.958					
Ho	8.374				8.70	
Er	8.452					
Tm	8.271					
Yb	8.114					
Lu	8.108					
Hf	8.291				8.63	
Ta	7.975	8.04	8.05	8.09	7.98	8.03
W	7.692	7.72		7.71	7.71	7.73
Re	7.340	7.35				7.36
Os	7.080					
Ir	7.067	7.08		7.10	7.10	7.05
Pt	7.330	7.35	7.33	7.37	7.38	7.37
Au	7.743		7.70	7.73	7.72	7.69
Hg	8.507				8.82	
Tl	9.637					
Pb	10.565			10.40	10.55	
Bi	10.890				11.61	
Po	12.705					
At	11.812					
Rn	13.049					
Fr	12.725					
Ra	12.456					
Ac	11.499					
Th	10.651		10.84			

TABLE 4

The DFT-calculated and experimental ²³⁰ P = 0 atomic volumes of the six heavy elements (in m³/Mmol). The zero-point contribution (ZP) has been added to the FPLMTO in the fourth column.

Element	EMTO	FPLMTO	FPLMTO + ZP	Experiment
Polonium (sc)	23.33	22.40	22.43	22.97
Astatine (fcc)	23.42	22.90	22.94	---
Radon (fcc)	52.57	48.20	49.75	50.50
Francium (bcc)	61.98	64.40	64.45	---
Radium (bcc)	42.41	40.75	40.84	41.09
Actinium (fcc)	27.99	26.50	26.52	22.55

FIGURE CAPTIONS

FIGURE 1

Comparison of experimental and Debye-Grüneisen thermal expansion for selected elements.

FIGURE 2

Atomic volume vs. Atomic number for $1 \leq Z \leq 90$ and $P = 0$ GPa.

FIGURE 3

Atomic volume vs. Atomic number for $1 \leq Z \leq 90$ and $P = 1$ GPa.

FIGURE 4

Atomic volume vs. Atomic number for $1 \leq Z \leq 90$ and $P = 3$ GPa.

FIGURE 5

Atomic volume vs. Atomic number for $1 \leq Z \leq 90$ and $P = 10$ GPa.

FIGURE 6

Atomic volume vs. Atomic number for $1 \leq Z \leq 90$ and $P = 30$ GPa.

FIGURE 7

Atomic volume vs. Atomic number for $1 \leq Z \leq 90$ and $P = 100$ GPa.

FIGURE 8

Sketch of the enthalpy of reaction of various compounds under pressure. The $\Delta H = 0$ line is shown for reference.

FIGURE 1

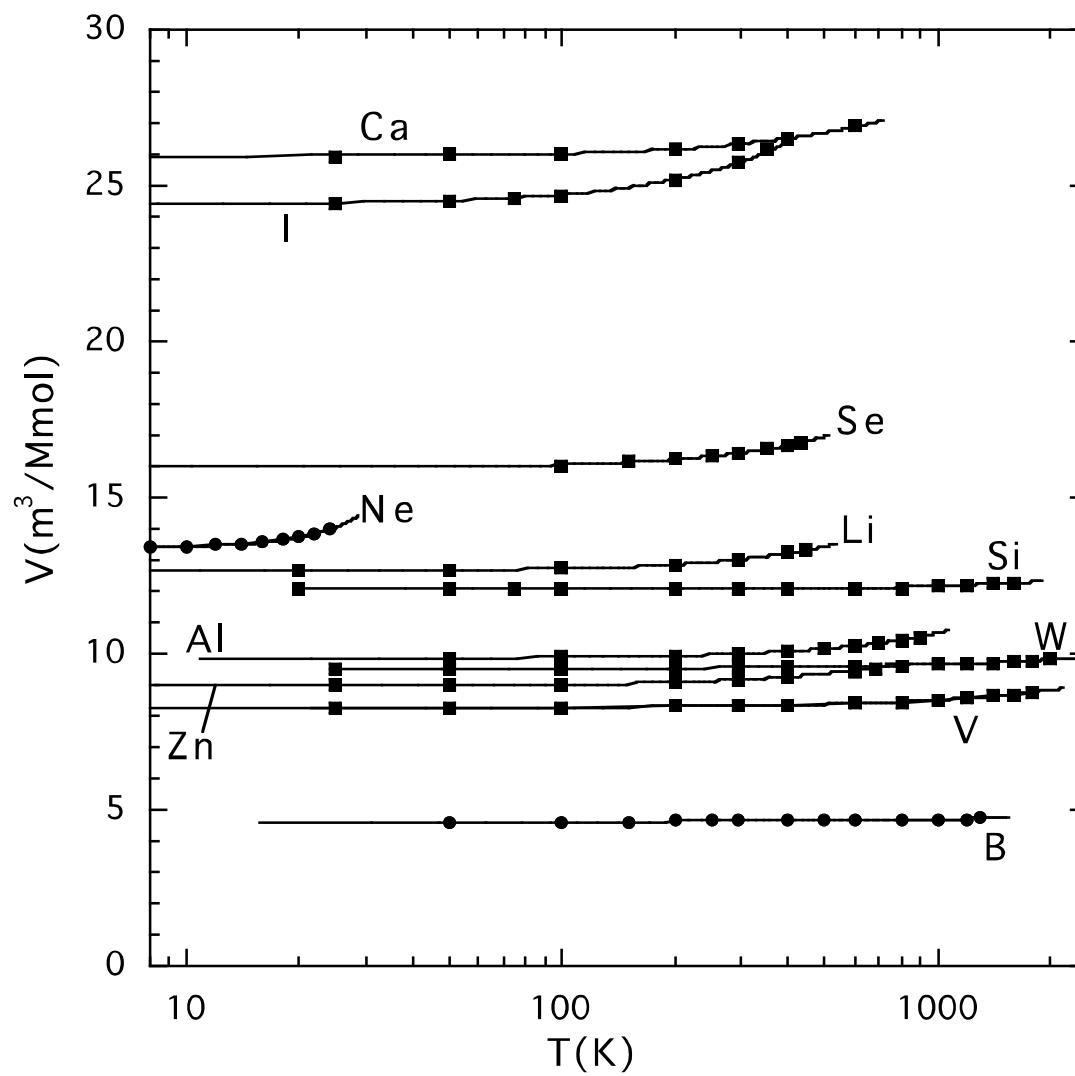


FIGURE 2

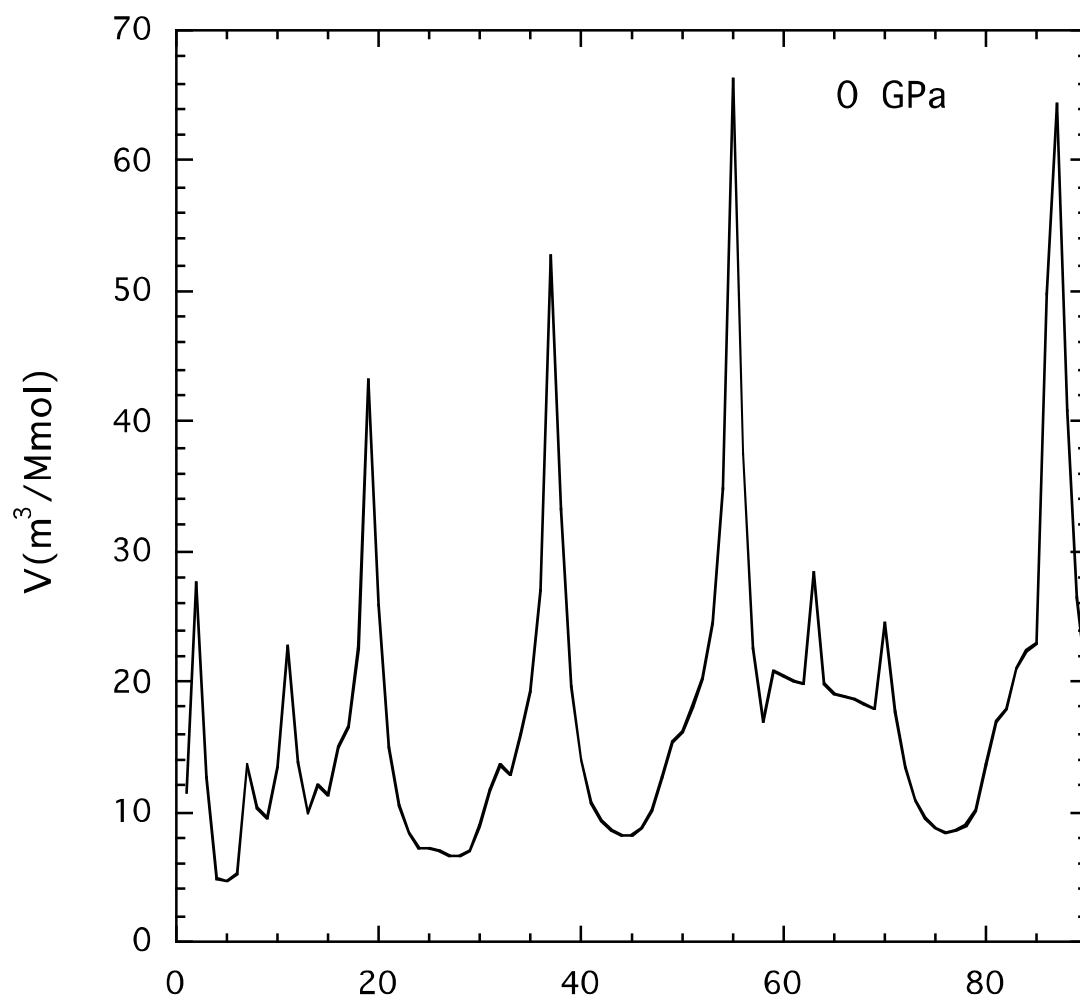


FIGURE 3

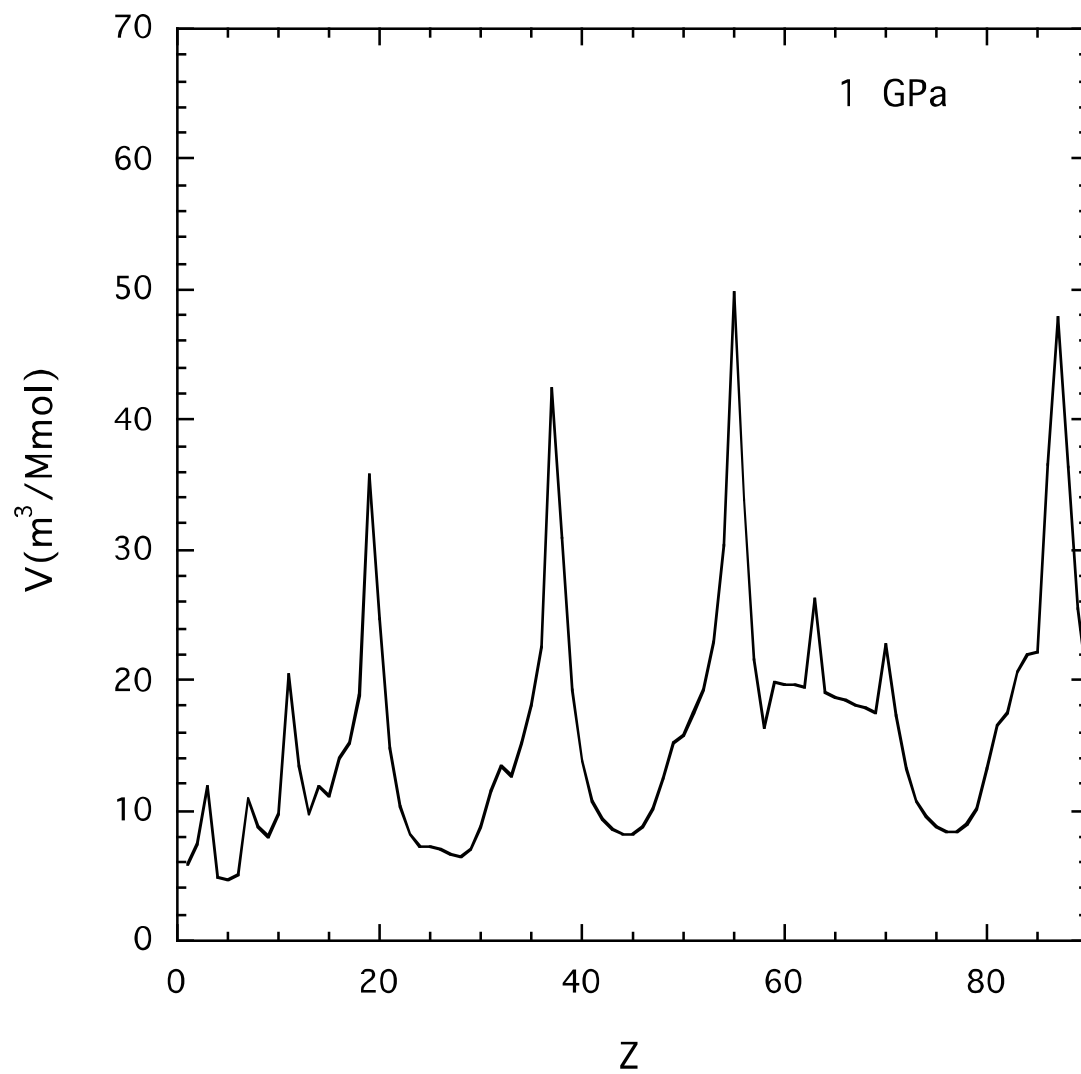


FIGURE 4

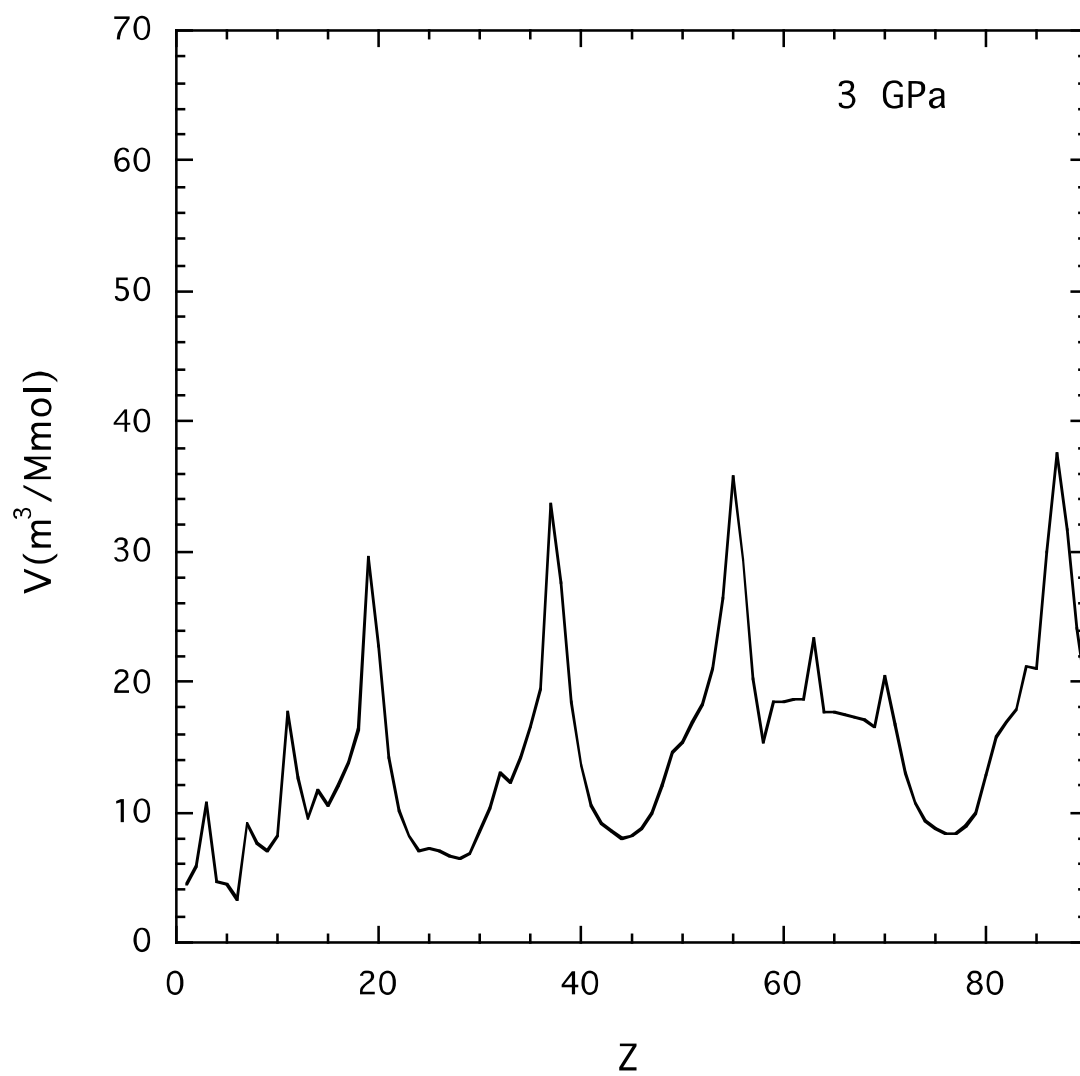


FIGURE 5

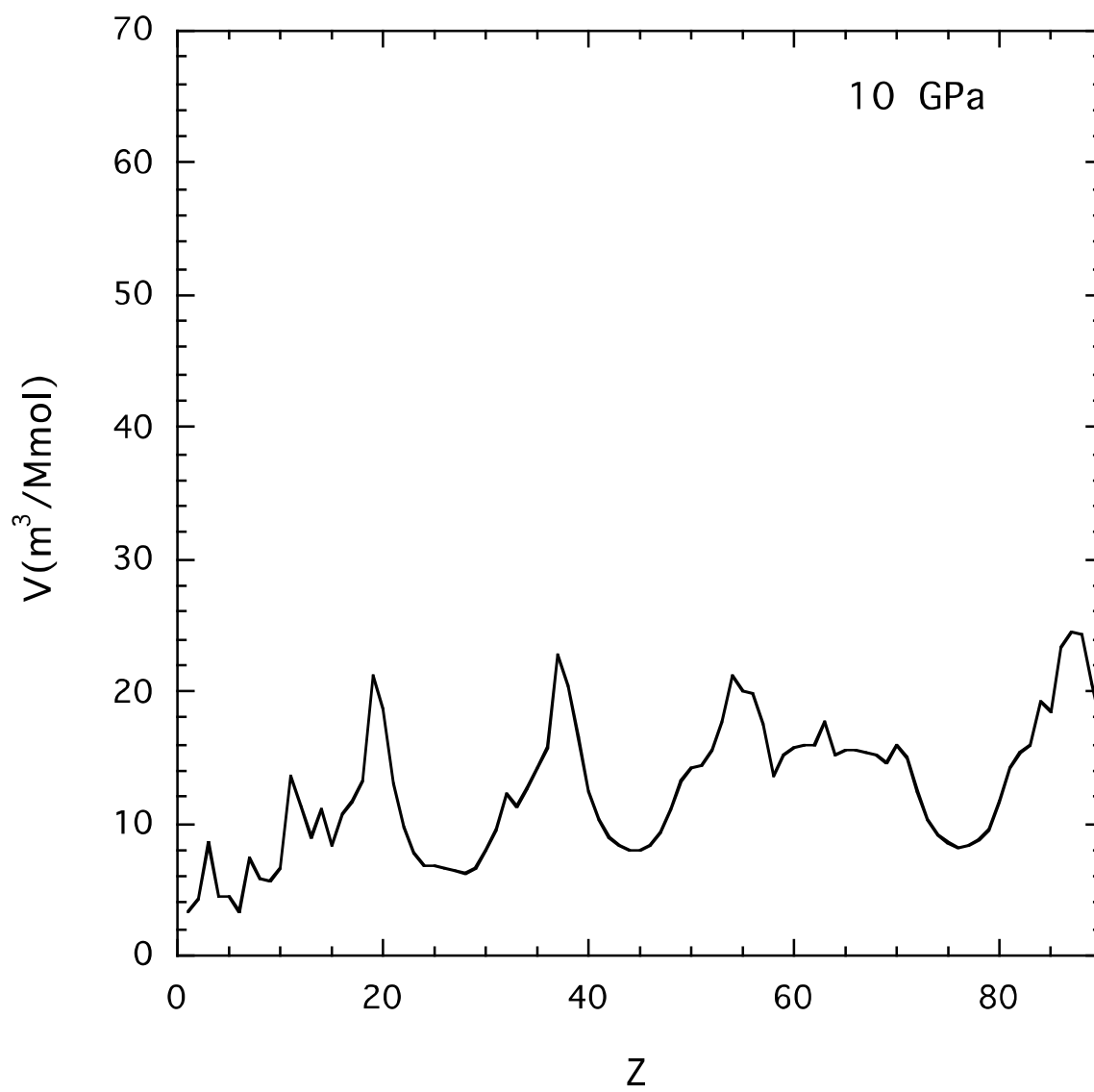


FIGURE 6

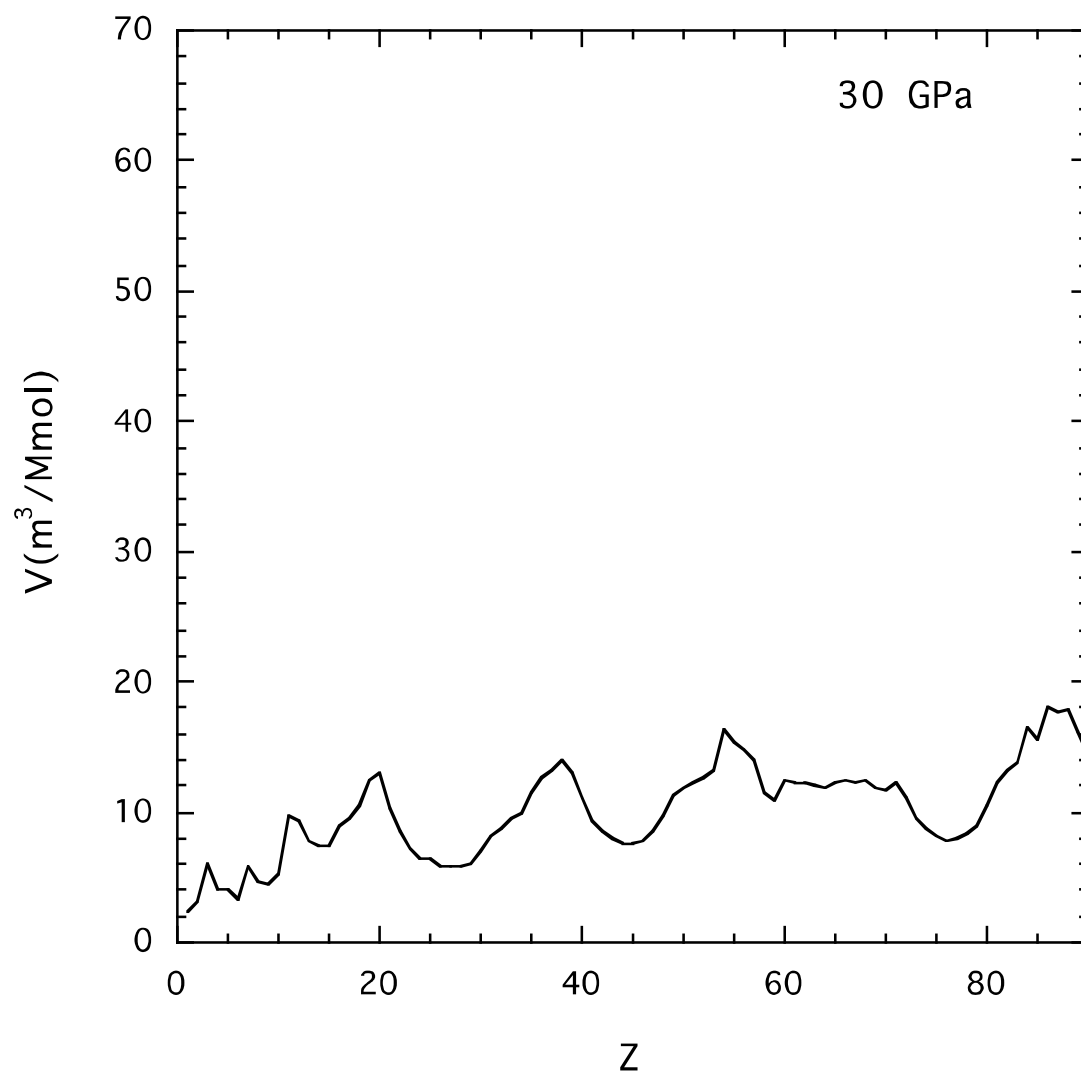


FIGURE 7

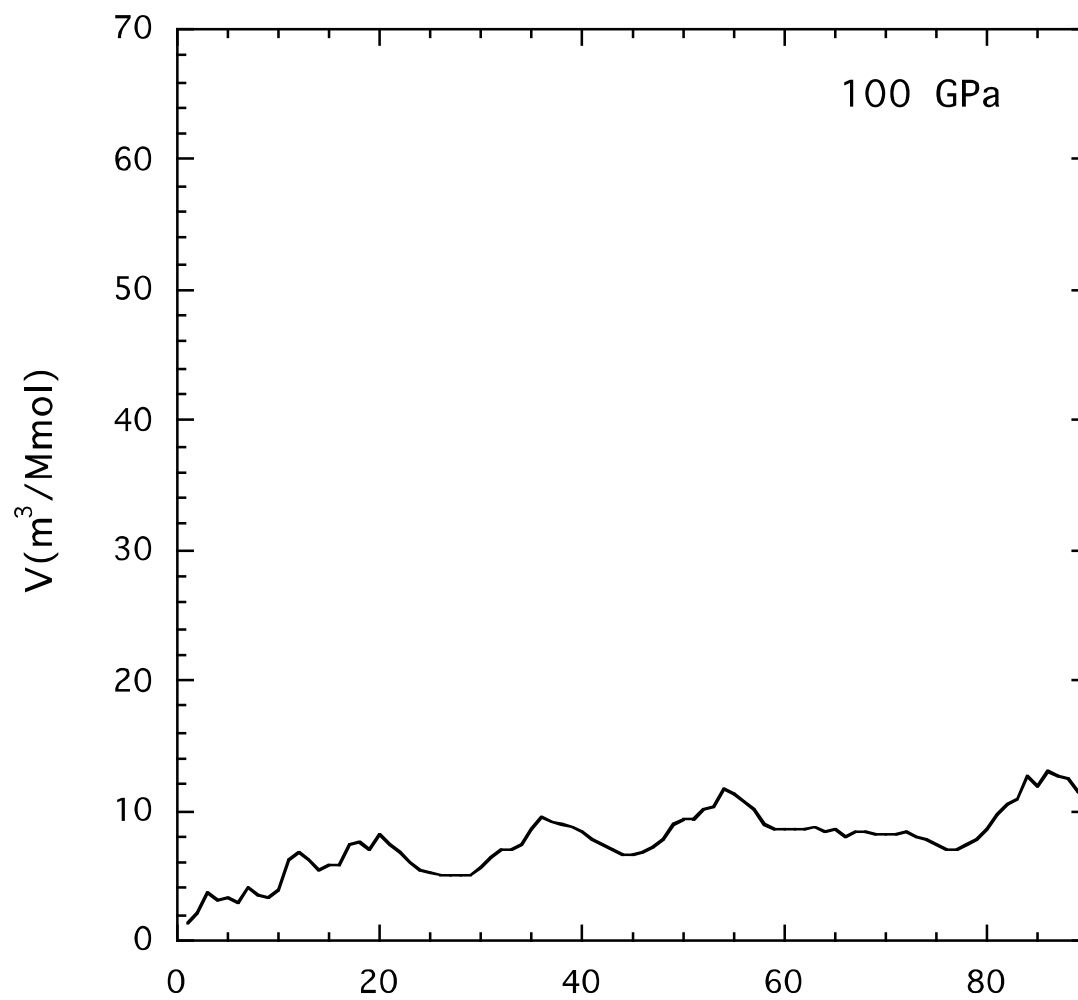


FIGURE 8

